

ARC-SPRUCE: Post-Quantum Blind Signatures and Anonymous Rate-Limited Credentials from Lattices

ABSTRACT

Blind signatures and anonymous credentials have recently seen a large interest from practitioners due to their usefulness in building privacy-preserving online authentication systems such as the EU Digital Identity Wallet. However, all currently known and practically efficient systems are not secure against quantum computers.

In this work, we use recent lattice-based techniques to modularly design post-quantum blind signatures and anonymous rate-limiting credentials. By using a state-of-the-art Non-Interactive Zero-Knowledge Proof system called SmallWood combined with a PRF based on the Poseidon Hash function, we obtain signature sizes that outperform existing constructions. In addition, our implementation shows competitive computational performance on modest hardware. We also provide the first, to our knowledge, implementation of the SmallWood proof system, which is of independent interest.

CCS CONCEPTS

• **Security and privacy** → **Digital signatures; Privacy-preserving protocols; Pseudonymity, anonymity and untraceability.**

KEYWORDS

Blind Signatures, Anonymous Credentials, Post-Quantum Cryptography

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1 INTRODUCTION

Anonymous credentials (AC) [Cha85, Bra95, CL01] are cryptographic protocols that allow parties to convey certified information about themselves in a privacy-preserving way. There are usually three roles in ACs, namely *issuers*, *holders* and *verifiers*. More concretely, an AC scheme allows a holder to prove to a verifier that it possesses certain attributes, which have been authenticated by the issuer. Here, the authenticated attributes are usually called a *credential*, which

is kept private by the user. Using a so-called *showing* process, the user can generate a presentation which convinces the verifier that attributes stored in its credential (or credentials) satisfy predicates without revealing any additional information. While there are numerous security properties that ACs can achieve, the most fundamental ones are unforgeability and unlinkability. Here, the former ensures that an issuer indeed generated a credential that a user presents, and that colluding issuers and verifiers may not link a presentation to a specific issuance session. The latter should also ensure that multiple presentations cannot be traced to the same credential.

Certain AC use-cases like rate limiting [YWFH26, SK25] require that the credential is only valid for a *bounded* number of presentations in a given context (e.g. per epoch or per service) – without revealing which presentations come from the same holder up to the enforced limit. A key advantage of existing, pre-quantum ACs is their performance, e.g. with constant credential and presentation size, regardless of the enforced rate limit. However, the situation looks very different in the post-quantum setting. There, practically efficient AC schemes [BLNS23a, AGJ⁺24, PFWF23], are yet mostly absent. This is highly unfortunate, given the uptake of (privacy-preserving) digital identification schemes in e.g. the EU Digital Identity Wallet, and simultaneous requirement for deployed schemes to eventually be quantum-secure: given current migration timelines, post-quantum ACs may become mandatory in the next 5 years or fewer. Recently, Google [AS26] moved the timeline for post-quantum to 2029. While everlasting anonymous rate-limited tokens [CADLT25] provide post-quantum privacy, they only guarantee rate limiting until quantum computers arrive.

Building ACs. To understand how post-quantum ACs (with rate-limiting) are usually constructed, it is instructive to understand how classically-secure constructions work. Anonymous credentials are often built on top of a Blind Signature design. Here, one of the most well-known approach (which is also the most promising in the post-quantum setting) is due to Fischlin [Fis06]:

- (1) The issuer initially creates a key pair (sk, vk) using an EUF CMA-secure signature scheme. We assume that user and verifier have vk , while the issuer keeps sk .
- (2) The holder first commits to its message μ as t , using randomness r . It then sends t to the issuer.
- (3) Upon receiving t , the issuer signs it and thus creates a signature sig . It then sends sig to the user, who checks its verifiability with vk .
- (4) To create the blind signature, the user creates a Non-Interactive Zero-Knowledge Proof of Knowledge (NIZK) showing that it knows sig, r such that

$$\text{Verify}_{vk}(t, sig) = 1 \wedge t = \text{Commit}(\mu; r)$$

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Combining existing post-quantum primitives generically, even with zero-knowledge friendly hash functions [PFW23] used in the signature scheme, leads to non-competitive performance. Thus, existing works [BLNS23b, AGJ⁺24, FR25a] carefully choose commitment schemes, signatures and NIZKs such that the NIZK for the given relation has a small proof size and prover/verifier time. However, the state-of-the-art is generally not considered efficient enough for practical deployments: Constructing blind signatures from alternative post-quantum hash-and-sign cryptography [BFM⁺25] and anonymous credentials from lattices [LCL⁺23], where recent work brings the credential size down to just over 70 kB using optimized samplers [JRS23, JS25].

1.1 Contributions

In this work, we narrow the efficiency gap for post-quantum blind signatures and ACs substantially. For Blind Signatures, we achieve this by optimizing existing design ideas carefully and employ novel primitives. We then extend our construction to yield the first post-quantum rate-limiting anonymous credentials. Crucially, the size of our credential as well as the presentation is essentially independent of the rate limit.

Our construction builds on proof-friendly lattice signatures from vanishing SIS (vSIS) with rational hashing [DKLW25a, DKLW25b] and on fast trapdoor generation and Gaussian preimage sampling for NTRU-like lattices [ENS⁺23]. For succinct zero knowledge over small fields, we use the SmallWood framework for hash-based polynomial commitments and compiled PIOPs [FR25b].

While SmallWood has been used for ZK-friendly post-quantum signatures in CAPSS [FR25a], we use a zk-friendly PRF with a significantly smaller modulus for tag derivation, which reduces the size of our communication significantly, and discuss Poseidon/Poseidon2-style instantiations [GKR⁺21, GKS23].

We construct a Blind Signature based on a signature whose security relies on lattice assumptions together with a hash-based NIZK. We then extend this to an *Anonymous Rate-Limited Credential* (ARC) [YWFH26] by including a Pseudorandom Function (PRF) into our design. Using a key committed in the message that is signed by the issuer, the user is able to evaluate a PRF on the key on different nonces. This PRF output, which we call a *tag*, can be made public. Since the signature and thus the key remain hidden due to the use of a NIZK, this intuitively makes the scheme unlinkable. By including a proof of correct PRF evaluation into the NIZK, the user also cannot generate a differing tag for the same nonce. Therefore, any two presentations generated for the same credential are linkable as they must have the same tag, which makes the scheme rate-limiting.

To design the Blind Signature, we follow the recent proposal of [BLNS23b, LSS24, DKLW25b] which also presents blind signatures and anonymous credentials based on lattice assumptions. We first let the holder commit to its message μ using an aligned linear commitment. Then, in an interactive process, issuer and signer sample the randomness that

determines a public target T to be signed by the issuer. Here, the user proves in zero knowledge that T is consistent with the committed message and the issuer-supplied inverse-side randomness. The signature scheme is formed by a matrix $\mathbf{A}$ together with a trapdoor $\mathbf{td}$ that allows sampling of short preimages $\text{mod } q$ using a GPV-style [GPV08] preimage sampler. The issuer then uses $\mathbf{td}$ to sample a short preimage $\mathbf{u}$ such that $\mathbf{A}\mathbf{u} = T \text{ mod } q$, which serves as the signature. To prove knowledge of a valid $\mathbf{u}$ we then use the SmallWood NIZK scheme [FR25b]. While this approach makes the round complexity of issuance non-optimal (requiring 3 rounds instead of 1 round), it allows tighter parameters for the preimage sampler while making the statement proven in the NIZK structurally simple. To turn this construction into an ARC, we let the user additionally commit to a PRF key $\mathbf{k}$ for the Poseidon2 hash function [GKS23]. It turns out that the parameters such as the modulus q of Poseidon2 can be chosen to be compatible with those of the lattice commitment, signature scheme and NIZK, making the proof of PRF evaluation highly efficient. The holder can then attach an evaluation of the PRF on a nonce where the committed key is used, together with a proof of correct PRF evaluation, as part of the presentation. If the holder reuses the nonce, this will be noticed by a verifier keeping track of previously issued tags. To the best of our knowledge, classical rate-limiting constructions also require a verifier keeping a state. [CDVW24, YWFH26, SK25]

1.2 Technical overview

This work constructs ARC around a single protocol blueprint with full signed message $\mu = \mathbf{m}||\mathbf{k}$, where $\mathbf{m}$ is a vector of attributes and $\mathbf{k}$ is a *secret* PRF key.

Issuance. Let $\mathbf{A}_\mu = [\mathbf{A}_m \mid \mathbf{A}_k]$ and $\mathbf{A}_s$ be the public matrices of the aligned linear commitment. In the issuance protocol, the holder samples a short attribute vector $\mathbf{m}$, a secret PRF key $\mathbf{k}$, forms the full message $\mu = \mathbf{m}||\mathbf{k}$, and additionally samples linear-side randomness $R_0 = (S||E)$ to compute the commitment

$$\mathbf{c} = \mathbf{A}_m\mathbf{m} + \mathbf{A}_k\mathbf{k} + \mathbf{A}_sS + E$$

that is sent to the issuer. The issuer, in response, samples inverse-side randomness R_1 such that $B_3 - R_1 \in R_q^\times$ and returns R_1 to the holder. The holder then computes

$$Z = (B_3 - R_1)^{-1}, \quad T = B_0 + U\mathbf{c} + Z = h_{\text{tran}, \mu, (R_0, R_1)}(B),$$

and sends T to the issuer together with a NIZK that it is well-formed according to $\mathbf{c}$ and R_1 and that all involved values are short.

The signature scheme is based on a hash-and-sign signature. It precomputes preimages of a matrix $\mathbf{A}$ using a trapdoor Gaussian preimage sampler over an NTRU-like lattice (Antrag/hybrid sampling) to instantiate the issuer's trapdoor signing step. The short preimage $\mathbf{u}$ is then computed by the issuer and returned to the holder, who verifies that $T = \mathbf{A}\mathbf{u}$.

We illustrate the issuance transcript flow and pre-sign proofs in Figure 1.

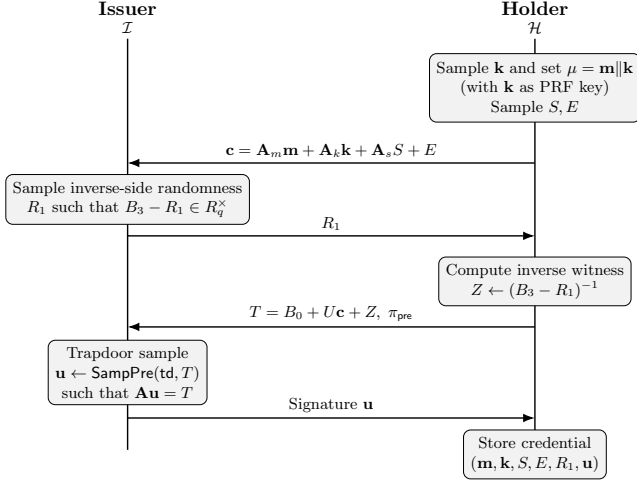


Figure 1: Issuance flow (ARC). The issuance proof π_{issue} binds T to the commitment and the inverse-side randomness, while hiding $\mathbf{m}, \mathbf{k}$.

Blind Signatures and ARC Presentations. Let $R_0 = (S||E)$. A presentation is a NIZK showing knowledge of $\mathbf{m}, \mathbf{k}, R_0, R_1, Z, T, \mathbf{u}$ where $\mu = \mathbf{m}||\mathbf{k}$, such that $T = h_{\text{tran}, \mu, (R_0, R_1)}(B)$ and $T = \mathbf{A}\mathbf{u}$, with all hidden values short. Toward constructing ARC, we additionally compute $\text{tag} = F(\mathbf{k}, \text{nonce})$, append it to the NIZK, and prove that nonce is in the rate-limiting interval and that tag is well-formed.

The verifier checks the NIZK proof and enforces the rate policy by maintaining server-side state keyed by $(\text{nonce}, \text{tag})$ (or an application-specific encoding thereof) to avoid double-spending.

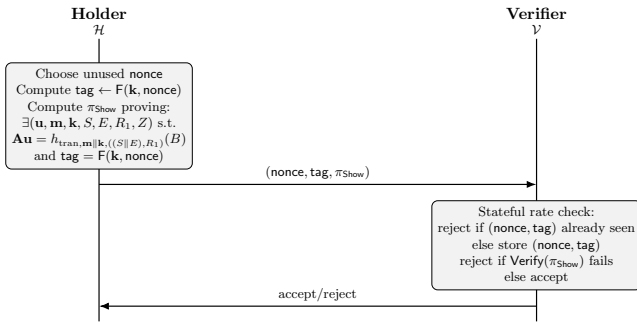


Figure 2: Showing (ARC presentation). Using an unused nonce $\leq \text{presentation_limit}$, the holder computes a deterministic tag from $(\mathbf{k}, \text{nonce})$ (rate limiting) and pseudorandom across unused nonces (unlinkability).

Figure 2 summarizes how tags enable stateful rate limiting without revealing the credential.

SmallWood Proofs. SmallWood [FR25b] is a hash-based polynomial commitment scheme which produces efficient witnesses for polynomials of small degree $\leq 2^{16}$, which is

particularly useful for lattice-based problems. The construction avoids the overhead of asymptotically-designed systems like STARKS while achieving better concrete efficiency than MPC-in-the-head variants for small witnesses. SmallWood exclusively relies on hash-based primitives, making it plausibly post-quantum secure. We use SmallWood as an NIZK for polynomial constraints over a small field $\mathbb{F}_q$, and we treat it as a black box that provides: (i) commitment to witness rows packed on an evaluation domain, (ii) succinct openings at verifier-chosen points, and (iii) soundness and zero knowledge after Fiat–Shamir [FR25b].

To obtain compact proofs, both issuance and showing are compiled into SmallWood statements over committed witness rows, with verification of the aligned commitment, the inverse-witness relation, the rational-hash relation, signature verification, and PRF evaluation. The exact compiled relations are specified in Section 5, while proof-system internals are deferred to Appendix C.

In the showing proof, we certify in the replay space that a full transform-basis replay image vanishes. The theorem suite in Section 5 then transports that replay-space statement back to the coefficient-domain BB-tran identity.

2 PRELIMINARIES

This section defines notation and gives interface-level definitions for the lattice primitives, the NIZK model, the blind-signature syntax, ARC syntax and security goals, and pseudorandom functions.

2.1 Notation

Let λ denote the security parameter and $\text{negl}(\lambda)$ be any positive function negligible in λ . We write PPT for probabilistic algorithms whose runtime is polynomial in λ . For integers $a < b$ we write $[a, b] := \{a, a+1, \dots, b\}$ and we use $[b]$ as shorthand for $[1, b]$. Throughout this work, we denote vectors in lower-case bold letters such as $\mathbf{b}$ and matrices in upper-case bold letters such as $\mathbf{A}$.

We work over the cyclotomic ring $R = \mathbb{Z}[X]/(X^N + 1)$ for power-of-two N and its quotient $R_q = R/qR$, identified with polynomials of degree $< N$ with coefficients in $\mathbb{Z}_q$. For $a \in \mathbb{Z}$, we write $\langle a \rangle_q \in \{0, \dots, q-1\}$ for reduction modulo q . When considering $\mathbb{Z}_q$ -elements from an interval $[-a, b]$, we identify these with their representatives in $\{-q/2, \dots, q/2\}$. For elements $x \in R_q$ we write $\|x\| \leq B$ to mean that each coefficient of x comes from $[-B, B]$. For vectors $\mathbf{x} \in R_q^\ell$, $\|\mathbf{x}\| \leq B$ means that the statement holds for all elements of $\mathbf{x}$.

All our constructions share the same modulus q , ring R_q , and coefficient bound β_{msg} . We write $\mathcal{M}$ for the bounded message space appropriate to the primitive under discussion. In the blind-signature and ARC layers, the full signed message has length $\ell_\mu = \ell_m + \ell_k$ and is written as $\mu = (\mathbf{m}||\mathbf{k})$. We assume that all public dimension and modulus parameters are collected in pp and honestly generated by $\text{Setup}(1^\lambda)$. All other setup algorithms for primitives obtain $1^\lambda, \text{pp}$ as input, and we do not mention this explicitly. Whenever we use the

CRT decomposition $R_q \cong \prod_i F_i$, we write

$$(x_i)_i \odot (y_i)_i := (x_i y_i)_i, \quad (x_i)_i \oslash (y_i)_i := (x_i y_i^{-1})_i$$

whenever every $y_i \neq 0$. Thus $\odot$ and $\oslash$ denote slotwise multiplication and, where defined, slotwise division in the CRT basis.

2.2 Commitment schemes and aligned linear commitments

Definition 2.1 (Commitment scheme). A (non-interactive) commitment scheme is a tuple of PPT algorithms

$$\text{Com} = (\text{Setup}, \text{Commit}, \text{Verify})$$

with the following syntax:

- $\text{pp}_{\text{Com}} \leftarrow \text{Setup}(1^\lambda, \text{pp})$ is a PPT algorithm which outputs public parameters pp_{Com} . It is implicit input to all other algorithms.
- $c \leftarrow \text{Commit}(m)$ is a PPT algorithm which takes a message $m \in \mathcal{M}$ as input and outputs a commitment c and decommitment d .
- $\text{Verify}(c, m, d) \in \{0, 1\}$ deterministically checks validity of the opening.

Correctness holds if for all $\text{pp}_{\text{Com}} \leftarrow \text{Setup}$, $m \in \mathcal{M}$ and $(c, d) \leftarrow \text{Commit}(m)$ it holds that

$$\text{Verify}(c, m, d) = 1.$$

The scheme is *computationally binding* if for all PPT adversaries $\mathcal{A}$,

$$\Pr \left[\text{pp}_{\text{Com}} \leftarrow \text{Setup}(1^\lambda, \text{pp}) \wedge (c, m, d, m', d') \leftarrow \mathcal{A}(\text{pp}_{\text{Com}}) \wedge m \neq m' \wedge \text{Verify}(c, m, d) = 1 \wedge \text{Verify}(c, m', d') = 1 \right] \leq \text{negl}(\lambda),$$

where the probability is taken over the randomness of Setup , $\mathcal{A}$.

The scheme is (*computationally*) *hiding* if for all PPT distinguishers $\mathcal{D}$, all messages $m_0, m_1 \in \mathcal{M}$ and $\text{pp}_{\text{Com}} \leftarrow \text{Setup}(1^\lambda, \text{pp})$,

$$\left| \Pr[\mathcal{D}^{\mathcal{O}_{\text{Com}}(m_0)}(\text{pp}_{\text{Com}}) = 1] - \Pr[\mathcal{D}^{\mathcal{O}_{\text{Com}}(m_1)}(\text{pp}_{\text{Com}}) = 1] \right| \leq \text{negl}(\lambda)$$

where $\mathcal{O}_{\text{Com}}(m_b)$ runs $(c, d) \leftarrow \text{Commit}(m_b)$ and outputs c and the probability is taken over the randomness of Commit , $\mathcal{D}$.

2.2.1 Aligned linear commitment. We use a bounded linear commitment over R_q whose opening is split into a message part and a linear-side randomness pair. Fix message length ℓ_μ , linear-secret dimension ℓ_s , output length ℓ_{Com} , coefficient bound β_{msg} , and bounded-support sampling distributions χ_s, χ_e supported on coefficient intervals inside $[-\beta_{\text{msg}}, \beta_{\text{msg}}]$. The public commitment key is written as

$$\mathbf{A}_{\text{Com}} = [\mathbf{A}_\mu \mid \mathbf{A}_s \mid I_{\ell_{\text{Com}}}],$$

where $\mathbf{A}_\mu \xleftarrow{\$} R_q^{\ell_{\text{Com}} \times \ell_\mu}$ and $\mathbf{A}_s \xleftarrow{\$} R_q^{\ell_{\text{Com}} \times \ell_s}$.

- $\text{Setup}(1^\lambda, \text{pp})$ samples $\mathbf{A}_\mu, \mathbf{A}_s$ as above and outputs $\text{pp}_{\text{Com}} := (\mathbf{A}_\mu, \mathbf{A}_s, \beta_{\text{msg}})$.
- The message space is

$$\mathcal{M} := \{\mu \in R_q^{\ell_\mu} \mid \|\mu\|_\infty \leq \beta_{\text{msg}}\}.$$

- $\text{Commit}(\mu)$ samples $S \leftarrow \chi_s^{\ell_s}$ and $E \leftarrow \chi_e^{\ell_{\text{Com}}}$ and outputs

$$\mathbf{c} = \mathbf{A}_\mu \mu + \mathbf{A}_s S + E$$

together with opening (S, E) .

- $\text{Verify}(\mathbf{c}, \mu, (S, E))$ accepts iff

$$\mathbf{c} \in R_q^{\ell_{\text{Com}}}, \quad \mu \in \mathcal{M}, \quad \|S\|_\infty, \|E\|_\infty \leq \beta_{\text{msg}},$$

$$\text{and } \mathbf{c} = \mathbf{A}_\mu \mu + \mathbf{A}_s S + E.$$

Definition 2.2 (Module-SIS and Module-ISIS over R_q). Fix integers $n, m \geq 1$, modulus $q \geq 2$, and bound $\beta > 0$.

- $\mathbf{M} - \text{SIS}_\infty(n, m, q, \beta)$. Given $\mathbf{A} \in R_q^{n \times m}$, find a non-zero $\mathbf{s} \in R^m$ such that $\mathbf{A}\mathbf{s} \equiv 0 \pmod{q}$ and $\|\mathbf{s}\|_\infty \leq \beta$.
- $\mathbf{M} - \text{ISIS}_\infty(n, m, q, \beta)$. Given $(\mathbf{A}, \mathbf{t}) \in R_q^{n \times m} \times R_q^n$, find $\mathbf{s} \in R^m$ such that $\mathbf{A}\mathbf{s} \equiv \mathbf{t} \pmod{q}$ and $\|\mathbf{s}\|_\infty \leq \beta$.

Lemma 2.3 (Binding of the aligned commitment). *Let $\mathcal{A}$ be any PPT adversary that breaks the binding property of the aligned commitment scheme with probability p for random matrices $\mathbf{A}_\mu, \mathbf{A}_s$ and coefficient bound β_{msg} . Then there exists a PPT attacker $\mathcal{A}'$ that breaks the $\text{MSIS}_\infty(n, m, q, 2\beta_{\text{msg}})$ assumption with essentially the same probability and runtime.*

PROOF. If two distinct openings $(\mu, S, E) \neq (\mu', S', E')$ verify the same commitment, then

$$\mathbf{A}_\mu(\mu - \mu') + \mathbf{A}_s(S - S') + (E - E') = 0$$

in $R_q^{\ell_{\text{Com}}}$. The concatenated difference is non-zero and coefficient-wise bounded by $2\beta_{\text{msg}}$, hence yields an MSIS_∞ solution with essentially the same probability and runtime. $\square$

The hiding property of the aligned commitment is stated below together with the aligned h_{tran} target used by the issuance protocol.

2.3 Lattice trapdoors and rational hashing

We recall only the definitions needed to state our construction, where we follow the definitions [DKLW25a]. We defer additional definitions such as the vSIS - and GenISIS_f -assumptions to Appendix A.1.

Definition 2.4 (Lattice trapdoor suite). A lattice trapdoor suite over R_q consists of PPT algorithms $(\text{Setup}, \text{TrapGen}, \text{SampPre})$ such that:

- $(\text{pp}_{\text{td}}) \leftarrow \text{Setup}(1^\lambda, \text{pp})$ generates admissible public parameters $(n, m, q, \beta_{\text{sig}})$ contained in pp_{td} .
- $(\mathbf{A}, \text{td}) \leftarrow \text{TrapGen}(\text{pp}_{\text{td}})$ outputs a public matrix $\mathbf{A} \in R_q^{n \times m}$ and a trapdoor td , where the distribution of $\mathbf{A}$ is statistically close to uniform over $R_q^{n \times m}$.
- On input $(\text{pp}_{\text{td}}, \text{td}, \mathbf{t})$ with $\mathbf{t} \in R_q^n$, the algorithm $\text{SampPre}(\text{td}, \mathbf{t})$ outputs $\mathbf{u} \in R^m$ such that $\mathbf{A}\mathbf{u} = \mathbf{t}$ in R_q^n , except with negligible failure probability, and $\|\mathbf{u}\|_\infty \leq \beta_{\text{sig}}$ except with negligible probability.

We defer details of the sampler that we use in this work to Appendix B.

Definition 2.5 (Trapdoor rational hash). A trapdoor rational hash scheme over R_q is a tuple

$$\text{TDRH} = (\text{Setup}, \text{TrapGen}, \text{Eval}, \text{SampPre})$$

consisting of the lattice trapdoor suite above together with a public evaluation algorithm. We extend **Setup** so that pp_{td} also contains a public hash description B , a randomness space $\mathcal{X}$ and an admissibility error bound δ . For fixed B , the algorithm $\text{Eval}(\text{pp}_{\text{td}}, m, \chi)$ takes a message $m \in \mathcal{M}$ from the ambient message space fixed by pp and witness randomness $\chi \in \mathcal{X}$ and outputs a target $\mathbf{t} \in R_q^n$ or $\perp$ if the input is inadmissible. For a public key B contained in pp_{td} , define the admissible message subspace

$$\mathcal{M}_B := \left\{ m \in \mathcal{M} : \Pr_{\chi \leftarrow \mathcal{X}} [\text{Eval}(\text{pp}_{\text{td}}, m, \chi) = \perp] \leq \delta \right\}.$$

Correctness requires that for every $m \in \mathcal{M}_B$ and every $\chi \in \mathcal{X}$ with $\text{Eval}(\text{pp}_{\text{td}}, m, \chi) = \mathbf{t} \neq \perp$, the output $\mathbf{u} \leftarrow \text{SampPre}(\text{pp}_{\text{td}}, \mathbf{t}, \chi)$ satisfies $\mathbf{A}\mathbf{u} = \mathbf{t}$ and $\|\mathbf{u}\|_\infty \leq \beta_{\text{sig}}$, except with negligible probability.

This interface does not by itself require the public target $\mathbf{t}$ to hide m ; in our aligned instantiation, that privacy is supplied later by the hiding statements of Section 2.3.1.

In this work, we use the BB-tran-type rational hash defined in [DKLW25a, Table 1]. In the concrete ARC-SPRUCE instantiation below, we specialize the generic target space to the single-coordinate case $n = 1$.

Definition 2.6 (Rational hash, admissibility, and eliminated form). Our concrete instantiation uses a public description $B = (B_0, B_1, B_2, B_3)$, where $B_0, B_3 \in R_q$ and B_1, B_2 denote public linear maps into R_q . Let $m \in \mathcal{M}$, let R_0 denote the linear-side randomness, and let $R_1 \in R_q$ denote the inverse-side randomness. Write

$$L_B(m, R_0) := B_0 + B_1 m + B_2 R_0.$$

We say that (B, m, R_0, R_1) is *admissible* if $B_3 - R_1 \in R_q^\times$, equivalently, if $B_3 - R_1$ is non-zero in every CRT slot of R_q .

When this admissibility condition holds, the rational hash is the partial function

$$T = h_{\text{tran}, m, (R_0, R_1)}(B) \iff \exists Z \in R_q : T = L_B(m, R_0) + Z \wedge (B_3 - R_1) \odot Z = 1.$$

Equivalently, the eliminated relation is

$$(B_3 - R_1) \odot (T - L_B(m, R_0)) = 1. \quad (1)$$

Lemma 2.7 (Eliminated form under admissibility). *Let $R_q \cong \prod_{i=1}^s F_i$ be the CRT decomposition of R_q into field factors, and write*

$$L_B(m, R_0) = (\ell_1, \dots, \ell_s), \quad B_3 - R_1 = (d_1, \dots, d_s), \quad T = (t_1, \dots, t_s) \text{ with } (B_3 - R_1)^{-1} \in R_q, \quad \text{so that} \quad (B_3 - R_1) \odot Z = 1. \quad (2)$$

If (B, m, R_0, R_1) is admissible, that is, $d_i \neq 0$ for every i , then

$$T = h_{\text{tran}, m, (R_0, R_1)}(B) \iff (B_3 - R_1) \odot (T - L_B(m, R_0)) = 1.$$

PROOF. Under the CRT isomorphism, the defining relation is the coordinate-wise statement

$$t_i = \ell_i + d_i^{-1} \quad \text{for every } i.$$

If $d_i \neq 0$ for every i , then each d_i is invertible in F_i , so

$$t_i = \ell_i + d_i^{-1} \iff d_i(t_i - \ell_i) = 1.$$

Collecting the coordinates yields the equivalence. If admissibility fails, then $d_i = 0$ in at least one slot, so the quotient relation is undefined there and the eliminated relation is outside its intended scope. $\square$

Note that the undefined case $h_{\text{tran}, m, (R_0, R_1)}(B) = \perp$ is exactly the inadmissibility event that $B_3 - R_1$ is not invertible in R_q , and is modeled by the admissibility parameter δ .

2.3.1 Aligned commitment and aligned BB-tran target. For the blind-issuance surface used later, we parse the full signed message as

$$\mu = (\mathbf{m} \| \mathbf{k}) \in R_q^{\ell_\mu}, \quad \ell_\mu = \ell_m + \ell_k, \quad \mathbf{A}_\mu = [\mathbf{A}_m \mid \mathbf{A}_k].$$

Let the linear-side randomness be split as

$$R_0 := (S \| E), \quad S \in R_q^{\ell_s}, \quad E \in R_q^{\ell_{\text{com}}}, \quad R_0 \leftarrow \mathcal{D}_0 := \mathcal{X}_s^{\ell_s} \times \mathcal{X}_e^{\ell_{\text{com}}}.$$

Let the inverse-side randomness be sampled from an admissible distribution

$$R_1 \leftarrow \mathcal{D}_1^{\text{adm}}$$

supported on values satisfying $B_3 - R_1 \in R_q^\times$. Equivalently, one may sample from an unconditional source $\mathcal{D}_1$ and restart until admissible, incurring an inadmissibility term

$$\delta_{\text{adm}} := \Pr_{R_1 \leftarrow \mathcal{D}_1} [B_3 - R_1 \notin R_q^\times].$$

Define the aligned commitment by

$$c := \text{Commit}(\mu; R_0) := \mathbf{A}_m \mathbf{m} + \mathbf{A}_k \mathbf{k} + \mathbf{A}_s S + E. \quad (2)$$

Let $U \in R_q^{\ell_{\text{com}}}$ be a public linear map from the commitment space to R_q , and define the BB-tran blocks by

$$B_{1,m} := U \mathbf{A}_m, \quad B_{1,k} := U \mathbf{A}_k, \quad B_{2,s} := U \mathbf{A}_s, \quad B_{2,e} := U. \quad (3)$$

For later sections, we use the shorthand

$$B_1(M + K) := B_{1,m}M + B_{1,k}K, \quad B_2 R_0 := B_{2,s}S + B_{2,e}E.$$

Then the aligned BB-tran linear part is

$$L_B(\mathbf{m}, \mathbf{k}, S, E) := B_0 + B_{1,m} \mathbf{m} + B_{1,k} \mathbf{k} + B_{2,s} S + B_{2,e} E = B_0 + U c. \quad (4)$$

Define the inverse witness

$$Z := h_{\text{tran}, \mu, (R_0, R_1)}(B) := L_B(\mathbf{m}, \mathbf{k}, S, E) + Z. \quad (6)$$

Finally define the aligned target

$$T = B_0 + B_{1,m} \mathbf{m} + B_{1,k} \mathbf{k} + B_{2,s} S + B_{2,e} E + Z = B_0 + U c + Z. \quad (7)$$

Equivalently,

Hiding-view experiment. For a PPT adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$, define $\text{Exp}_{h_{\text{tran}}, \mathcal{A}}^{\text{hide-view}}(\lambda)$ as follows.

- (1) Sample $\mathbf{A}_m, \mathbf{A}_k, \mathbf{A}_s, U, B_0, B_3$ as above and give

$$pp_{\text{hid}} := (\mathbf{A}_m, \mathbf{A}_k, \mathbf{A}_s, U, B_0, B_3)$$

to $\mathcal{A}_1$.

- (2) $\mathcal{A}_1(pp_{\text{hid}})$ outputs two challenge messages

$$\mu_0 = (\mathbf{m}_0 \| \mathbf{k}_0), \quad \mu_1 = (\mathbf{m}_1 \| \mathbf{k}_1),$$

together with state st .

- (3) The challenger samples

$$b \xleftarrow{\$} \{0, 1\}, \quad (S, E) \leftarrow \mathcal{D}_0, \quad R_1 \leftarrow \mathcal{D}_1^{\text{adm}}.$$

- (4) It computes

$$c^* := \mathbf{A}_m \mathbf{m}_b + \mathbf{A}_k \mathbf{k}_b + \mathbf{A}_s S + E, \\ Z^* := (B_3 - R_1)^{-1},$$

and

$$T^* := B_0 + U c^* + Z^*.$$

- (5) It gives (c^*, T^*) to $\mathcal{A}_2$, which outputs a bit b' .

The experiment returns 1 iff $b' = b$. Define

$$\mathcal{A}_{h_{\text{tran}}}^{\text{hide-view}}(\mathcal{A}) := \left| \Pr \left[\text{Exp}_{h_{\text{tran}}, \mathcal{A}}^{\text{hide-view}}(\lambda) = 1 \right] - \frac{1}{2} \right|. \quad (8)$$

Target-only hiding experiment. Define $\text{Exp}_{h_{\text{tran}}, \mathcal{A}}^{\text{hide-target}}(\lambda)$ identically, except that the challenger gives only T^* to $\mathcal{A}_2$. Let

$$\mathcal{A}_{h_{\text{tran}}}^{\text{hide-target}}(\mathcal{A}) \quad (9)$$

denote the corresponding advantage.

Theorem 2.8 (Commitment-lifted hiding of aligned BB-tran). *Let Commit be the aligned commitment from (2). For every PPT adversary $\mathcal{A}$ against $\text{Exp}_{h_{\text{tran}}, \mathcal{A}}^{\text{hide-view}}$, there exists a PPT adversary $\mathcal{B}$ against hiding of Commit such that*

$$\mathcal{A}_{h_{\text{tran}}}^{\text{hide-view}}(\mathcal{A}) = \mathcal{A}_{\text{Com}}^{\text{hide}}(\mathcal{B}). \quad (10)$$

Consequently, if Commit is hiding, then the aligned BB-tran issuer view (c, T) is hiding.

PROOF. Construct $\mathcal{B} = (\mathcal{B}_1, \mathcal{B}_2)$ for the commitment-hiding game as follows.

$\mathcal{B}_1$ samples

$$U \xleftarrow{\$} R_q^{1 \times \ell_{\text{Com}}}, \quad B_0, B_3 \xleftarrow{\$} R_q,$$

and receives from the commitment challenger the public commitment parameters

$$pp_{\text{Com}} = (\mathbf{A}_m, \mathbf{A}_k, \mathbf{A}_s).$$

It forwards

$$pp_{\text{hid}} = (\mathbf{A}_m, \mathbf{A}_k, \mathbf{A}_s, U, B_0, B_3)$$

to $\mathcal{A}_1$. When $\mathcal{A}_1$ outputs

$$\mu_0 = (\mathbf{m}_0 \| \mathbf{k}_0), \quad \mu_1 = (\mathbf{m}_1 \| \mathbf{k}_1), \quad \text{st},$$

$\mathcal{B}_1$ forwards the same message pair to its commitment-hiding challenger.

Let c^* be the commitment challenge returned by that challenger, so by definition there is a hidden bit b such that

$$c^* = \mathbf{A}_m \mathbf{m}_b + \mathbf{A}_k \mathbf{k}_b + \mathbf{A}_s S + E$$

for $(S, E) \leftarrow \mathcal{D}_0$.

Now $\mathcal{B}_2$ samples

$$R_1 \leftarrow \mathcal{D}_1^{\text{adm}}, \quad Z^* := (B_3 - R_1)^{-1},$$

and computes

$$T^* := B_0 + U c^* + Z^*. \quad (11)$$

It gives (c^*, T^*) to $\mathcal{A}_2$ and outputs whatever bit b' $\mathcal{A}_2$ returns.

It remains to check that the distribution seen by $\mathcal{A}$ is exactly the real one. Using (3), (4), and the defining equation for c^* ,

$$\begin{aligned} T^* &= B_0 + U(\mathbf{A}_m \mathbf{m}_b + \mathbf{A}_k \mathbf{k}_b + \mathbf{A}_s S + E) + Z^* \\ &= B_0 + (U \mathbf{A}_m) \mathbf{m}_b + (U \mathbf{A}_k) \mathbf{k}_b + (U \mathbf{A}_s) S + U E + Z^* \\ &= B_0 + B_{1,m} \mathbf{m}_b + B_{1,k} \mathbf{k}_b + B_{2,s} S + B_{2,e} E + Z^* \\ &= h_{\text{tran}, \mu_b, ((S, E), R_1)}(B). \end{aligned} \quad (12)$$

So the pair (c^*, T^*) is distributed exactly as in $\text{Exp}_{h_{\text{tran}}, \mathcal{A}}^{\text{hide-view}}$. Therefore the advantages are equal, proving (10). $\square$

Theorem 2.9 (LWE-based hiding of aligned BB-tran). *Assume the decisional module-LWE problem for the distribution pair (χ_s, χ_e) is hard in the following form: given*

$$\mathbf{A}_s \xleftarrow{\$} R_q^{\ell_{\text{Com}} \times \ell_s}$$

and a challenge vector $Y \in R_q^{\ell_{\text{Com}}}$, distinguish between

$$Y = \mathbf{A}_s S + E \quad \text{with} \quad S \leftarrow \chi_s^{\ell_s}, \quad E \leftarrow \chi_e^{\ell_{\text{Com}}}, \quad (13)$$

and

$$Y \xleftarrow{\$} R_q^{\ell_{\text{Com}}}. \quad (14)$$

Then for every PPT adversary $\mathcal{A}$,

$$\mathcal{A}_{h_{\text{tran}}}^{\text{hide-view}}(\mathcal{A}) \leq \mathcal{A}_{q, \ell_s, \ell_{\text{Com}}, \chi_s, \chi_e}^{\text{d-MLWE}}(\mathcal{C}) \quad (15)$$

for some PPT distinguisher $\mathcal{C}$. In fact the reduction below is exact.

PROOF. Build a distinguisher $\mathcal{C}$ for decisional module-LWE. It receives a challenge pair $(\mathbf{A}_s, Y)$, where Y is either an LWE sample as in (13) or uniform as in (14).

First, $\mathcal{C}$ samples independently

$$\begin{aligned} \mathbf{A}_m &\xleftarrow{\$} R_q^{\ell_{\text{Com}} \times \ell_m}, & \mathbf{A}_k &\xleftarrow{\$} R_q^{\ell_{\text{Com}} \times \ell_k}, \\ U &\xleftarrow{\$} R_q^{1 \times \ell_{\text{Com}}}, & B_0, B_3 &\xleftarrow{\$} R_q. \end{aligned}$$

It gives

$$pp_{\text{hid}} = (\mathbf{A}_m, \mathbf{A}_k, \mathbf{A}_s, U, B_0, B_3)$$

to $\mathcal{A}_1$, which returns challenge messages

$$\mu_0 = (\mathbf{m}_0 \| \mathbf{k}_0), \quad \mu_1 = (\mathbf{m}_1 \| \mathbf{k}_1),$$

and state st .

Then $\mathcal{C}$ samples

$$b \xleftarrow{\$} \{0, 1\}, \quad R_1 \leftarrow \mathcal{D}_1^{\text{adm}}, \quad Z^* := (B_3 - R_1)^{-1},$$

and sets

$$c^* := \mathbf{A}_m \mathbf{m}_b + \mathbf{A}_k \mathbf{k}_b + Y, \quad (16)$$

$$T^* := B_0 + U c^* + Z^*. \quad (17)$$

It gives (c^*, T^*) to $\mathcal{A}_2$, receives a guess b' , and outputs 1 iff $b' = b$.

If $Y = \mathbf{A}_s S + E$ is an LWE sample, then

$$c^* = \mathbf{A}_m \mathbf{m}_b + \mathbf{A}_k \mathbf{k}_b + \mathbf{A}_s S + E,$$

hence by the same calculation as in (12),

$$T^* = h_{\text{tran}, \mu_b, ((S, E), R_1)}(B),$$

so (c^*, T^*) is exactly the real hiding-view challenge. Therefore

$$\Pr[\mathcal{C} \text{ outputs } 1 \mid Y = \mathbf{A}_s S + E] = \frac{1}{2} + \mathcal{A}_{h_{\text{tran}}}^{\text{hide-view}}(\mathcal{A}). \quad (18)$$

If instead $Y \xleftarrow{\$} R_q^{\ell_{\text{com}}}$, then for each fixed b ,

$$c^* = \mathbf{A}_m \mathbf{m}_b + \mathbf{A}_k \mathbf{k}_b + Y$$

is still uniform on $R_q^{\ell_{\text{com}}}$, because translating a uniform distribution by a fixed public offset preserves uniformity. Hence c^* is independent of b . Since R_1 is sampled independently of b , so are Z^* and

$$T^* = B_0 + U c^* + Z^*.$$

Thus the whole pair (c^*, T^*) is independent of the hidden bit b , and $\mathcal{A}_2$ has success probability exactly $1/2$:

$$\Pr[\mathcal{C} \text{ outputs } 1 \mid Y \xleftarrow{\$} R_q^{\ell_{\text{com}}}] = \frac{1}{2}. \quad (19)$$

Subtracting (19) from (18), the distinguishing advantage of $\mathcal{C}$ is exactly (15). $\square$

Corollary 2.10 (The rational-hash output itself is hiding). *For every PPT adversary $\mathcal{A}$ in the target-only game,*

$$\mathcal{A}_{h_{\text{tran}}}^{\text{hide-target}}(\mathcal{A}) \leq \mathcal{A}_{h_{\text{tran}}}^{\text{hide-view}}(\mathcal{A}') \leq \mathcal{A}^{\text{d-MLWE}}(\mathcal{C}) \quad (20)$$

for suitable PPT adversaries $\mathcal{A}', \mathcal{C}$.

PROOF. Given a target-only adversary $\mathcal{A}$, build a view adversary $\mathcal{A}'$ that simply ignores the commitment component c^* and passes only T^* to $\mathcal{A}$. This preserves the success probability exactly. Then apply Theorem 2.9. $\square$

Remark 2.11. The privacy statement established for issuance is issuer-view hiding of the public issuance surface. Full blindness in the two-session sense of Definition 2.19 is left as a separate problem.

The load-bearing public identity is

$$T = B_0 + U c + (B_3 - R_1)^{-1}.$$

The issuance syntax in Section 3.0.2 is written so that this identity holds literally.

We now define a signature scheme for a Trapdoor Rational Hash.

Definition 2.12 (vSIS-style hash-and-sign). Let TDRH be a trapdoor rational hash scheme with parameters pp_{td} . The associated signature scheme vSIS = (KG, Sign, Verify) is defined by

- $\text{KG}(1^\lambda)$ runs $(\mathbf{A}, \text{td}) \leftarrow \text{TrapGen}(\text{pp}_{\text{td}})$ and outputs $\text{vk} = \mathbf{A}$, $\text{sk} = \text{td}$.
- $\text{Sign}(\text{pp}_{\text{td}}, \text{sk}, m)$ for $m \in \mathcal{M}_B$ samples $\chi \leftarrow \mathcal{X}$ until $\mathbf{t} = \text{Eval}(B, m, \chi) \neq \perp$, computes $\mathbf{u} \leftarrow \text{SampPre}(\text{td}, \mathbf{t})$, and outputs $\sigma = (\mathbf{u}, \chi)$.

- $\text{Verify}(\text{pp}_{\text{td}}, \text{vk}, m, \sigma)$ for $\sigma = (\mathbf{u}, \chi)$ accepts iff $m \in \mathcal{M}_B$, $\chi \in \mathcal{X}$, $\text{Eval}(B, m, \chi) \neq \perp$, $\|\mathbf{u}\|_\infty \leq \beta_{\text{sig}}$, and

$$\mathbf{A}\mathbf{u} = \text{Eval}(B, m, \chi).$$

Correctness requires that all admissible parameters pp_{td} generated by $\text{Setup}()$, correctly generated (vk, sk) and $m \in \mathcal{M}_B$, the signatures generated using Sign are accepted by Verify except with negligible probability.

Security of vSIS-style signatures. We now define security as strong existential unforgeability under random-message attacks.

Definition 2.13 (Unforgeability for vSIS-style signatures). Let TDRH be a trapdoor rational hash scheme with parameters pp_{td} and vSIS = (KG, Sign, Verify) be the associated signature scheme. Fix $Q \in \text{poly}(\lambda)$. In the experiment $\text{Exp}_{\text{sEUUF-RMA}}^{\text{vSIS}, \mathcal{A}, Q}(\lambda)$, the challenger runs $(\text{vk}, \text{sk}) \leftarrow \text{KG}$, then uniformly samples $m_1, \dots, m_Q \leftarrow \mathcal{M}_B$ and computes $\sigma_i \leftarrow \text{Sign}(\text{sk}, m_i)$ for each $i \in [Q]$. Finally $\mathcal{A}$ on input $\text{vk}, (m_i, \sigma_i)_{i \in [Q]}$ and the trapdoor rational hash parameters outputs a pair (m^*, σ^*) and wins iff:

- (1) $\text{Verify}(\text{vk}, m^*, \sigma^*) = 1$, and
- (2) $(m^*, \sigma^*) \notin \{(m_i, \sigma_i) : 1 \leq i \leq Q\}$.

We can equivalently define sEUUF-CMA, sEUUF-SMA, and sSUF-SMA security where CMA-security allows the adversary to choose all queried messages after seeing vk , while SMA-security requires the adversary to fix them before seeing vk .

Lemma 2.14 (Imported security properties of the non-blind rational-hash signature layer). *Let $\ell_{\text{hash}} = 4$ for the present ARC instantiation and set*

$$k_{\text{DKLW}} := m + \ell_{\text{hash}}.$$

Assume the following:

- (1) pp_{td} are admissible parameters for (Setup, TrapGen, SampPre);
- (2) the inadmissibility bound satisfies $\delta \leq \text{negl}(\lambda)$;
- (3) a DKLW-compatible norm bound β_{DKLW} is fixed so that every signature accepted by the present verifier also satisfies $\|\mathbf{u}\| \leq \beta_{\text{DKLW}}$ in the norm used in [DKLW25a];
- (4) the predicate condition of [DKLW25a, Definition 7] holds for the instantiated rational-hash family; and
- (5) the denominator-bound, strong-linear-independence, and adaptive-security side conditions required by [DKLW25a, Theorems 4–8, Corollary 1, and Section 5.2] hold for the instantiated family.

Here β_{DKLW} is the verifier norm parameter from the imported DKLW theorems. It need not literally equal β_{sig} , but it must dominate every signature accepted by our verifier when measured in the norm convention of [DKLW25a]. Assumptions (4) and (5) are precisely the instantiated BB-tran compatibility conditions needed to invoke the cited DKLW reduction chain. Then the following security consequences hold.

- (1) The non-blind template of Definition 2.12 is correct.

- (2) Under the strong random-hinted vSIS assumption for the induced parameter tuple, the non-blind signature scheme is sEUF-RMA-secure.
- (3) For $f(B, m, \chi) = h_{\text{tran}, m, \chi}(B)$, the experiment GenISIS_f is equivalent to the sEUF-RMA experiment.
- (4) For the BB-tran-type instantiation used here, the GenISIS_f / IntGenISIS_f framework of [DKLW25a, Section 5.2 and Theorem 6] gives the corresponding adaptive/CMA security statement for the non-blind signature layer under the stated parameter conditions.
- (5) Under the strong-hinted vSIS assumption, the same non-blind family also enjoys sEUF-SMA security.

PROOF. Items (1), (2), (3), and (5) follow from [DKLW25a, Theorems 4, 5, 7, 8, and Corollary 1] under the stated specialization. Item (4) is the BB-tran-specific adaptive-security lift of [DKLW25a, Section 5.2 and Theorem 6], which places the non-blind signature layer on the $\text{GenISIS}_f/\text{IntGenISIS}_f$ interface used there for sEUF-CMA security. $\square$

Remark 2.15 (Scope of the DKLW import). Lemma 2.14 is the external reduction basis for the non-blind signature layer. It does not by itself imply blindness or one-more unforgeability of the blind issuance protocol, and it does not by itself yield an ARC security theorem.

2.4 Non-interactive zero-knowledge arguments of knowledge (NIZKs)

We use the NIZK-ARK syntax of the SmallWood framework instantiated in the random-oracle model [FR25b, Definition 3].

Definition 2.16 (NIZKs in the random oracle model). Let $\mathcal{R} \subseteq \mathcal{X} \times \mathcal{W}$ be a family of relations. A (transparent) non-interactive zero-knowledge *argument of knowledge* for $\mathcal{R}$ in the random oracle model (with oracle $\text{RO}(\cdot)$) is a pair of algorithms ($\text{Prove}^{\text{RO}}, \text{Verify}^{\text{RO}}$):

- $\text{Prove}^{\text{RO}}(w, x) \rightarrow \pi$ is probabilistic and on input (x, w) with $(x, w) \in R$ outputs a proof π .
- $\text{Verify}^{\text{RO}}(\pi, x) \rightarrow \{0, 1\}$ is a deterministic polynomial-time algorithm.

Completeness. For all $(x, w) \in \mathcal{R}$,

$$\Pr[\text{Verify}^{\text{RO}}(\pi, x) = 1 : \pi \leftarrow \text{Prove}^{\text{RO}}(w, x)] = 1.$$

Straightline-extractable knowledge soundness. There exists a PPT algorithm Ext , also called *extractor*, such that for any PPT adversary $\mathcal{A}$,

$$\Pr[\text{Verify}^{\text{RO}}(\pi, x) = 1 \wedge (x, w) \notin R] \leq \varepsilon,$$

where $(\pi, Q) \leftarrow \mathcal{A}^{\text{RO}}(x)$, Q is the set of query-response pairs made to RO by $\mathcal{A}$, and $w \leftarrow \text{Ext}(Q, \pi)$.

Zero knowledge. There exists a PPT algorithm Sim , called the *simulator*, such that for any $(x, w) \in \mathcal{R}$ and PPT algorithm $\mathcal{A}$:

$$\left| \Pr[\mathcal{A}^{\text{RO}}(\pi_{\text{real}}) = 1] - \Pr[\mathcal{A}^{\text{RO}}(\pi_{\text{sim}}) = 1] \right| \leq \xi,$$

where $\pi_{\text{real}} \leftarrow \text{Prove}^{\text{RO}}(w, x)$, $\pi_{\text{sim}} \leftarrow \text{Sim}^{\text{RO}}(x)$ and Sim can program the output of RO on any input.

We instantiate these proofs using the SmallWood PACS/PIOP framework compiled with Fiat-Shamir in the random oracle model [FR25b]. SmallWood is a commit-and-prove proof system that only relies on symmetric primitives. While its proof size is linear in $|w|$ it is sublinear in the circuit size x .

2.5 Blind signatures

A blind signature algorithm allows a holder $\mathcal{H}$ to obtain signatures valid under the verification key of an issuer $\mathcal{I}$, without the latter learning the signed message μ . Our definitions follow [Fis06]. When any algorithm $\mathcal{B}$ uses the interactive protocol Issue with $\mathcal{I}$, we will write $\mathcal{B}^{\text{Issue}(\cdot)}$ and adjust this the notation for other interactions accordingly.

Definition 2.17 (Blind Signature). A Blind Signature scheme is a tuple of algorithms

$$\text{BSig} = (\text{Setup}, \text{IssuerKeyGen}, \text{Issue}, \text{Verify})$$

such that:

- $\text{Setup}(1^\lambda)$ is a PPT algorithm that outputs public parameters pp_{BSig} . All other algorithms obtain $1^\lambda, \text{pp}_{\text{BSig}}$ as implicit inputs.
- $\text{IssuerKeyGen}()$ is a PPT algorithm that outputs issuer keys (vk, sk) .
- Issue is an interactive protocol between $\mathcal{I}$ and $\mathcal{H}$, where $\mathcal{I}$ has input sk and $\mathcal{H}$ has input μ, vk . At the end of the protocol, $\mathcal{H}$ obtains sig .
- $\text{Verify}(\text{vk}, \mu, \text{sig})$ is a deterministic polynomial-time algorithm that outputs a bit.

Definition 2.18 (Correctness). A blind signature scheme is correct if there exists negl such that for all messages μ ,

$$\Pr \left[\begin{array}{l} \text{pp}_{\text{BSig}} \leftarrow \text{Setup}(1^\lambda); \\ (\text{vk}, \text{sk}) \leftarrow \text{IssuerKeyGen}(\text{pp}_{\text{BSig}}); \\ \text{sig} \leftarrow \mathcal{H}^{\text{Issue}(\cdot)}(\mu, \text{vk}); \\ \text{Verify}(\text{vk}, \mu, \text{sig}) = 1 \end{array} \right] \geq 1 - \text{negl}(\lambda).$$

Definition 2.19 (Blindness). Let BSig be a blind signature scheme and let $\mu_0 \neq \mu_1$ be any two messages. Define the experiment $\text{Exp}_{\text{BSig}, \mathcal{A}}^{\text{sig-blind}}(\lambda)$ between a challenger $\mathcal{C}$ and a three-stage stateful adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3)$ as follows.

- (1) $\mathcal{C}$ runs $\text{pp}_{\text{BSig}} \leftarrow \text{Setup}(1^\lambda)$.
- (2) $\mathcal{A}_1(\text{pp}_{\text{BSig}})$ outputs a verification key vk .
- (3) Run $\mathcal{A}_2^{\text{Issue}(\cdot: \mathcal{H}(\mu_0, \text{vk})), \text{Issue}(\cdot: \mathcal{H}(\mu_1, \text{vk}))}$; at the end of which $\mathcal{C}$ obtains $\text{sig}_0, \text{sig}_1$.
- (4) If $\perp \in \{\text{sig}_0, \text{sig}_1\}$ then output a uniform bit and halt.
- (5) Otherwise, $\mathcal{C}$ samples $b \xleftarrow{\$} \{0, 1\}$ and gives the ordered pair $(\text{sig}_b, \text{sig}_{1-b})$ to $\mathcal{A}_3$, which outputs a bit b' .

Define $\mathcal{A}_{\text{BSig}}^{\text{sig-blind}}(\mathcal{A}) := \left| \Pr[b' = b] - \frac{1}{2} \right|$. We say BSig is *blind* if for all PPT $\mathcal{A}$ this advantage is negligible in λ .

Definition 2.20 (One-more unforgeability). A blind signature scheme is one-more unforgeable if there exists negl such

that for any PPT adversary $\mathcal{A}^{\text{Issue}(\cdot, \mathcal{I}(\text{sk}) \cdot \cdot)}(\text{vk})$ that makes at most $Q = Q(\lambda)$ oracle queries to **Issue**,

$$\Pr \left[\begin{array}{l} \text{pp}_{\text{BSig}} \leftarrow \text{Setup}(1^\lambda); \\ (\text{vk}, \text{sk}) \leftarrow \text{IssuerKeyGen}(\text{pp}_{\text{BSig}}); \\ \{(\mu_i, \text{sig}_i)\}_{i \in [Q+1]} \leftarrow \mathcal{A}^{\text{Issue}(\cdot, \mathcal{I}(\text{sk}) \cdot \cdot)}(\text{vk}) : \\ \forall 1 \leq i < j \leq Q+1 : \mu_i \neq \mu_j \wedge \\ \forall i \in [Q+1] : \text{Verify}(\text{vk}, \mu_i, \text{sig}_i) = 1 \end{array} \right] \leq \text{negl}(\lambda).$$

Rate limiting at the ARC layer. We do not introduce a separate stand-alone rate-limiting primitive. In the present paper, rate limiting is part of the ARC syntax itself: the holder presents a credential together with a public nonce and a PRF-derived public tag, and the verifier enforces the policy through local state on previously accepted $(\text{nonce}, \text{tag})$ pairs.

2.6 Anonymous Rate-Limiting Credentials (ARCs)

Anonymous rate-limited credentials [YWFH26] extend blind signatures by adding a showing algorithm **Show** that may create multiple presentations of a previously issued credential. We explicitly introduce the verifier party $\mathcal{V}$, which receives and checks those presentations. Our definition follows the established syntax and security models of [FHS19, BLNS23b], specialized to the ARC setting considered here.

We keep **Setup** and **IssuerKeyGen** separate, so the public commitment matrix is sampled independently of the issuer keys. We also separate **Show** and **Verify** to distinguish the client-side presentation algorithm from the verifier-side stateful check.

Definition 2.21 (ARC scheme syntax). Let $\mathcal{N}$ be a finite set of nonces. An anonymous rate-limited credential (ARC) scheme is a tuple of algorithms

$$\text{ARC} = (\text{Setup}, \text{IssuerKeyGen}, \text{Issue}, \text{Show}, \text{Verify})$$

such that:

- **Setup**(1^λ) is a PPT algorithm that outputs public parameters pp_{ARC} . All other algorithms obtain $1^\lambda, \text{pp}_{\text{ARC}}$ as implicit inputs.
- **IssuerKeyGen**() is a PPT algorithm that outputs issuer keys (vk, sk) .
- **Issue** is an interactive protocol between $\mathcal{I}$ and $\mathcal{H}$, where $\mathcal{I}$ has input sk and $\mathcal{H}$ has input a message μ and verification key vk . At the end of the protocol, $\mathcal{H}$ obtains the credential cred .
- **Show**($\text{cred}, \text{nonce}$) on input a credential cred and a nonce $\text{nonce} \in \mathcal{N}$ outputs a presentation pres .
- **Verify**($\text{vk}, \text{pres}, \text{state}$) is a deterministic polynomial-time algorithm that on input a verification key vk , a presentation pres , and a verifier state state outputs a validity bit together with an updated state state' .

Adjusting the blind-signature correctness notion of Definition 2.18 to ARC is immediate. Privacy at the ARC layer is captured below by presentation unlinkability, while issuance-side privacy continues to be expressed by Definition 2.19 together with the issuer-view hiding property proved in Section 3.

We additionally require the following soundness notion, which captures that each issuance can support at most one accepting presentation per nonce.

Definition 2.22 (ARC soundness). Let ARC be an ARC scheme with nonce space $\mathcal{N}$. For $Q = Q(\lambda)$, set

$$K := Q \cdot |\mathcal{N}| + 1.$$

Consider the following experiment. The challenger samples

$$\text{pp}_{\text{ARC}} \leftarrow \text{Setup}(1^\lambda), \quad (\text{vk}, \text{sk}) \leftarrow \text{IssuerKeyGen}(\text{pp}_{\text{ARC}}),$$

initializes the verifier state $\text{state}_0 = \emptyset$, and gives $(\text{pp}_{\text{ARC}}, \text{vk})$ to a PPT adversary $\mathcal{A}^{\text{Issue}(\cdot, \mathcal{I}(\text{sk}) \cdot \cdot)}$ that makes at most Q issuance queries. The adversary then outputs a sequence $(\text{pres}_i)_{i \in [K]}$. For $i = 1, \dots, K$, the challenger computes

$$(b_i, \text{state}_i) \leftarrow \text{Verify}(\text{vk}, \text{pres}_i, \text{state}_{i-1}).$$

We say that ARC is *sound* if for every PPT adversary, the probability that $b_i = 1$ for every $i \in [K]$ is negligible in λ .

This definition is the ARC analogue of one-more unforgeability. Since each honestly issued credential can be shown for all $|\mathcal{N}|$ different nonces, Q issuance interactions may legitimately yield $Q \cdot |\mathcal{N}|$ accepting presentations. Producing strictly more than that forces a policy violation for at least one nonce or credential by the pigeonhole principle.

However, we additionally require that if the same credential cred is presented multiple times, but for different nonces from $\mathcal{N}$, then the presentations should be unlinkable. To formalize this, we let a challenger create two credentials in interactions where the adversary controls the issuer. Then the challenger either creates two presentations for the same credential, or for different credentials.

Definition 2.23 (Presentation unlinkability). Let ARC be an ARC scheme. Consider the following experiment $\text{Exp}_{\text{ARC}, \mathcal{A}}^{\text{pres-unlink}}(\lambda)$ between a challenger $\mathcal{C}$ and a three-stage stateful adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3)$:

- (1) $\mathcal{C}$ runs $\text{pp}_{\text{ARC}} \leftarrow \text{Setup}(1^\lambda)$.
- (2) Run $(\text{vk}, \mu) \leftarrow \mathcal{A}_1(\text{pp}_{\text{ARC}})$.
- (3) Run $\mathcal{A}_2^{\text{Issue}(\cdot, \mathcal{C}(\text{vk}, \mu))}, \text{Issue}(\cdot, \mathcal{C}(\text{vk}, \mu))$, at the end of which $\mathcal{C}$ obtains $\text{cred}_0, \text{cred}_1$ and $\mathcal{A}_2$ outputs $\text{nonce}_0, \text{nonce}_1$.
- (4) If cred_0 or cred_1 is $\perp$ or if $\text{nonce}_0 = \text{nonce}_1$ then output a random bit.
- (5) $\mathcal{C}$ samples $b \xleftarrow{\$} \{0, 1, 2\}$ and computes

$$\begin{aligned} \text{pres}_0 &\leftarrow \text{Show}(\text{cred}_0, \text{nonce}_0) \\ \text{pres}_1 &\leftarrow \begin{cases} \text{Show}(\text{cred}_0, \text{nonce}_1) & \text{if } b = 0 \\ \text{Show}(\text{cred}_1, \text{nonce}_0) & \text{if } b = 1 \\ \text{Show}(\text{cred}_1, \text{nonce}_1) & \text{if } b = 2 \end{cases} \end{aligned}$$

- (6) $\mathcal{A}_3(\text{pres}_0, \text{pres}_1)$ outputs b' .

Define $\mathcal{A}_{\text{ARC}}^{\text{pres-unlink}}(\mathcal{A}) := |\Pr[b' = b] - \frac{1}{3}|$. We say ARC has *presentation unlinkability* if for all PPT $\mathcal{A}$ this advantage is negligible in λ .

2.7 Pseudorandom functions

Definition 2.24 (Pseudorandom function (PRF)). Let $\mathcal{K}$, $\mathcal{D}$ and $\mathcal{T}$ be sets such that $|\mathcal{K}|$ is exponential in λ . A function family $F : \mathcal{K} \times \mathcal{D} \rightarrow \mathcal{T}$ is a PRF if for all PPT adversaries $\mathcal{A}$,

$$A_F^{\text{prf}}(\mathcal{A}) := \left| \Pr[\mathcal{A}^{\mathcal{O}_0} = 1] - \Pr[\mathcal{A}^{\mathcal{O}_1} = 1] \right| \leq \text{negl}(\lambda),$$

where $\mathcal{O}_0(\cdot) = F(k, \cdot)$ for uniform $k \xleftarrow{\$} \mathcal{K}$ and $\mathcal{O}_1(\cdot)$ is a uniform random function from $\mathcal{D}$ to $\mathcal{T}$.

In this work, we instantiate PRFs based on the Poseidon2 construction [GKS23].

It is based on the Poseidon2 permutation, which is defined as follows:

Definition 2.25 (Poseidon2-style permutation structure). Let q be a prime, $d \geq 3$ such that $\gcd(d, q-1) = 1$ and $t, R_F, R_P \in \mathbb{N}$ where R_F is even. We call R_F the *external* rounds and R_P the *internal* rounds: external rounds apply the S-box to every lane before mixing, whereas internal rounds apply it only to the first lane. Furthermore, let $\{c_j^{(i)}\}_{j \in [R_F]} \subset \mathbb{F}_q$, $\{\hat{c}_i^{(i)}\}_{i \in [R_P]} \subset \mathbb{F}_q$ be public round constants and $\mathbf{M}_E, \mathbf{M}_I \in \mathbb{F}_q^{t \times t}$ be invertible maps.

For $\mathbf{x} = (x_1, \dots, x_t) \in \mathbb{F}_q^t$, define external rounds

$$E_i(\mathbf{x}) := \mathbf{M}_E \cdot ((x_1 + c_1^{(i)})^d, \dots, (x_t + c_t^{(i)})^d)^\top,$$

and internal rounds

$$I_i(\mathbf{x}) := \mathbf{M}_I \cdot ((x_1 + \hat{c}_1^{(i)})^d, x_2, \dots, x_t)^\top.$$

Then

$$P := E_{R_F} \circ \dots \circ E_{R_F/2+1} \circ I_{R_P} \circ \dots \circ I_1 \circ E_{R_F/2} \circ \dots \circ E_1.$$

We additionally define $\text{Tr}_i : \mathbb{F}_q^t \rightarrow \mathbb{F}_q^i$ outputs the first i elements of the input vector.

Definition 2.26 (Poseidon2 PRF). Let $\ell_{\text{tag}}, \ell_{\text{key}}, \ell_{\text{in}} \in \mathbb{N}$ and set $t = \ell_{\text{key}} + \ell_{\text{in}}$. For a key $\mathbf{k} \in \mathbb{F}_q^{\ell_{\text{key}}}$ and public input $\mathbf{n} \in \mathbb{F}_q^{\ell_{\text{in}}}$, we define the Poseidon2 PRF as

$$F(\mathbf{k}, \mathbf{n}) := \text{Tr}_{\ell_{\text{tag}}}(P(\mathbf{k} \parallel \mathbf{n}) + (\mathbf{k} \parallel \mathbf{n})),$$

where $P : \mathbb{F}_q^t \rightarrow \mathbb{F}_q^t$ is the permutation from Definition 2.25.

3 BLIND SIGNATURE CONSTRUCTION

This section presents the vSIS-based blind signature that underlies ARC-SPRUCE. It combines the aligned linear commitment of Section 2.2, the trapdoor rational-hash interface of Definition 2.5 instantiated with the concrete BB-tran form of Definition 2.6, and the NIZK-ARK of Definition 2.16. The signed message is denoted as μ .

3.0.1 Randomizing the rational-hash witness. In the protocol, $\mathcal{H}$ first commits to

$$(\mu, R_0) = (\mu, (S \parallel E)),$$

where R_0 is the linear-side randomness used by the aligned BB-tran target. Concretely, the holder samples S and E from the commitment-randomness distribution and computes

$$\mathbf{c} = \mathbf{A}_\mu \mu + \mathbf{A}_s S + E.$$

The issuer then samples admissible inverse-side randomness R_1 , and the holder derives the inverse witness

$$Z = (B_3 - R_1)^{-1}$$

so that the public target is

$$T = B_0 + U\mathbf{c} + Z = h_{\text{tran}, \mu, (R_0, R_1)}(B).$$

The issuing proof therefore binds one hidden opening witness simultaneously to the commitment and to the public target.

3.0.2 Blind Signature protocol. We assume $\text{NIZK} = (\text{Prove}, \text{Verify})$ is a NIZK for the relations below, $\text{Com} = (\text{Setup}, \text{Commit}, \text{Verify})$ is the aligned linear commitment, and

$$\text{TDRH} = (\text{Setup}, \text{TrapGen}, \text{Eval}, \text{SampPre})$$

is a trapdoor rational hash according to Definition 2.5. The public setup samples the aligned commitment key together with a public map U and public elements B_0, B_3 ; writing

$$B_1 := U\mathbf{A}_\mu, \quad B_2(S \parallel E) := (U\mathbf{A}_s)S + UE,$$

this defines the concrete BB-tran target used in issuance and verification.

We define the relations

$$\mathcal{R}_{\text{Issue, BS}} := \left\{ \left((\text{pp}_{\text{Com}}, \mathbf{c}, T, R_1, U, B), \begin{bmatrix} \mu \\ S \\ E \\ Z \end{bmatrix} \right) \mid \begin{array}{l} \text{pp}_{\text{Com}} = (\mathbf{A}_\mu, \mathbf{A}_s, \beta_{\text{msg}}) \wedge \\ \mathbf{c} = \mathbf{A}_\mu \mu + \mathbf{A}_s S + E \wedge \\ \mu \in \mathcal{M} \wedge \\ \|\mu, S, E\|_\infty \leq \beta_{\text{msg}} \wedge \\ R_1 \in \text{Supp}(\mathcal{D}_1^{\text{adm}}) \wedge \\ (B_3 - R_1) \odot Z = 1 \wedge \\ T = B_0 + U\mathbf{c} + Z \end{array} \right\}$$

and

$$\mathcal{R}_{\text{Sign}} := \left\{ \left((\text{pp}_{\text{Com}}, \mu, \mathbf{A}, U, B, \beta_{\text{msg}}, \beta_{\text{sig}}), \begin{bmatrix} S \\ E \\ R_1 \\ Z \\ \mathbf{u} \end{bmatrix} \right) \mid \begin{array}{l} \text{pp}_{\text{Com}} = (\mathbf{A}_\mu, \mathbf{A}_s, \beta_{\text{msg}}) \wedge \\ \mu \in \mathcal{M} \wedge \\ \|\mu, S, E\|_\infty \leq \beta_{\text{msg}} \wedge \\ \|\mathbf{u}\|_\infty \leq \beta_{\text{sig}} \wedge \\ R_1 \in \text{Supp}(\mathcal{D}_1^{\text{adm}}) \wedge \\ (B_3 - R_1) \odot Z = 1 \wedge \\ \mathbf{A}\mathbf{u} = B_0 + U(\mathbf{A}_\mu \mu + \mathbf{A}_s S + E) \end{array} \right\}$$

Our Blind Signature scheme is then defined as follows:

Setup(1^λ):

(1) Run

$$(\mathbf{A}_\mu, \mathbf{A}_s, \beta_{\text{msg}}) \leftarrow \text{Com.Setup}(1^\lambda, \text{pp}).$$

(2) Sample

$$U \xleftarrow{\$} R_q^{1 \times \ell_{\text{Com}}}, \quad B_0, B_3 \xleftarrow{\$} R_q,$$

and define the BB-tran description $B = (B_0, B_1, B_2, B_3)$ by

$$B_1 := U\mathbf{A}_\mu, \quad B_2(S \parallel E) := (U\mathbf{A}_s)S + UE.$$

(3) Set

$$\text{pp}_{\text{Com}} := (\mathbf{A}_\mu, \mathbf{A}_s, \beta_{\text{msg}}), \quad \mathcal{M} := \{\mu \in R_q^{\ell_\mu} : \|\mu\|_\infty \leq \beta_{\text{msg}}\},$$

$$\mathcal{X} := \mathcal{D}_0 \times \mathcal{D}_1^{\text{adm}}, \quad \delta := \delta_{\text{adm}}.$$

$$\text{pp}_{\text{td}} := (B, \mathcal{M}, \mathcal{X}, \delta, \beta_{\text{sig}}).$$

(4) Output

$$\text{pp}_{\text{BSig}} = (\text{pp}_{\text{Com}}, U, \text{pp}_{\text{td}}, \beta_{\text{msg}}).$$

IssuerKeyGen():

- (1) Compute $(\mathbf{A}, \text{td}) \leftarrow \text{TDRH.TrapGen}(\text{pp}_{\text{td}})$.
- (2) Output $\text{vk} = (\mathbf{A})$ and $\text{sk} = (\text{td})$.

Issue: $\mathcal{I}$ holds sk while $\mathcal{H}$ has μ, vk .

- (1) $\mathcal{H}$ samples $S \leftarrow \chi_s^{\ell_s}$ and $E \leftarrow \chi_e^{\ell_{\text{com}}}$.
- (2) $\mathcal{H}$ computes $\mathbf{c} \leftarrow \mathbf{A}_\mu \mu + \mathbf{A}_s S + E$ and sends $\mathbf{c}$ to $\mathcal{I}$.
- (3) $\mathcal{I}$ samples $R_1 \leftarrow \mathcal{D}_1^{\text{adm}}$ and sends R_1 to $\mathcal{H}$.
- (4) $\mathcal{H}$ sets $Z \leftarrow (B_3 - R_1)^{-1}$ and

$$T \leftarrow B_0 + U\mathbf{c} + Z = h_{\text{tran}, \mu, ((S\|E), R_1)}(B), \quad (21)$$

and produces the issuing NIZK π_{Issue} using **NIZK.Prove** proving knowledge of (μ, S, E, Z) satisfying relation $\mathcal{R}_{\text{Issue}, \text{BS}}$.

- (5) $\mathcal{H}$ sends (T, π_{Issue}) to $\mathcal{I}$.
- (6) $\mathcal{I}$ verifies π_{Issue} for relation $\mathcal{R}_{\text{Issue}, \text{BS}}$ on $(\text{pp}_{\text{Com}}, \mathbf{c}, T, R_1, U, B)$ using **NIZK.Verify**; if it accepts, sample $\mathbf{u} \leftarrow \text{TDRH.SampPre}(\text{td}, \mathbf{c})$ and send $\mathbf{u}$ to $\mathcal{H}$. Otherwise abort.
- (7) $\mathcal{H}$ checks $\mathbf{A}\mathbf{u} = T$ and that $\|\mathbf{u}\|_\infty \leq \beta_{\text{sig}}$. If not, abort.
- (8) Set $\text{sig} := (\mathbf{u}, S, E, R_1)$.

Verify($\text{vk}, \mu, \text{sig}$): For $\text{sig} = (\mathbf{u}, S, E, R_1)$:

- (1) Compute $\mathbf{c} \leftarrow \mathbf{A}_\mu \mu + \mathbf{A}_s S + E$.
- (2) Reject if $\|\mu, S, E\|_\infty > \beta_{\text{msg}}$ or $\|\mathbf{u}\|_\infty > \beta_{\text{sig}}$.
- (3) Reject if $B_3 - R_1 \notin R_q^\times$.
- (4) Compute $Z \leftarrow (B_3 - R_1)^{-1}$ and accept iff

$$\mathbf{A}\mathbf{u} = B_0 + U\mathbf{c} + Z.$$

3.0.3 Security.

Proposition 3.1 (Correctness). *Assume correctness of NIZK, TDRH. Then the construction is correct according to Definition 2.18.*

PROOF. The holder computes $T = B_0 + U\mathbf{c} + Z$ with $(B_3 - R_1) \odot Z = 1$, so the issuing relation is satisfiable by construction. Completeness of the pre-sign NIZK implies that $\mathcal{I}$ accepts the proof π_{Issue} for relation $\mathcal{R}_{\text{Issue}, \text{BS}}$. By correctness of **SampPre**, the returned $\mathbf{u}$ satisfies $\mathbf{A}\mathbf{u} = T$ and has norm $\leq \beta_{\text{sig}}$ except with negligible probability. The output $\text{sig} = (\mathbf{u}, S, E, R_1)$ therefore verifies except with negligible probability. $\square$

Proposition 3.2 (Issuer-view message hiding of issuance). *Assume Theorem 2.9 and that NIZK is zero knowledge. Then the public issuance surface*

$$(\mathbf{c}, R_1, T, \pi_{\text{Issue}})$$

is computationally hiding with respect to the holder message μ .

PROOF. We use two hybrids. In the first hybrid, replace the real issuing proof π_{Issue} by the simulated proof generated by the zero-knowledge simulator of NIZK. This changes the issuer view only negligibly. In the second hybrid, conditioned on the same public parameters and on the public issuer sample R_1 , the remaining holder-to-issuer view is exactly the pair $(\mathbf{c}, T)$ from the aligned BB-tran hiding experiment of Section 2.3.1, together with the independent public value R_1 . Theorem 2.9 therefore implies that $(\mathbf{c}, T)$ is computationally hiding as a function of μ , and adding the independent public

value R_1 preserves that hiding. Combining the two hybrids proves the claim. $\square$

Remark 3.3 (Security status of the blind protocol). The non-blind hash-and-preimage signature layer underlying this protocol is justified by the DKLW25 reduction summarized in Lemma 2.14. The issuance surface above additionally satisfies the issuer-view hiding property of Proposition 3.2. A proof of one-more unforgeability would still have to combine extractability of the pre-sign proof with an interactive hardness statement that genuinely matches the issuance oracle. Likewise, a full blindness theorem in the sense of Definition 2.19 would still have to strengthen the issuer-view simulation argument to the complete two-session experiment with correlated challenge signatures. Accordingly, the formal claims established here are non-blind signature security for the hash-and-preimage layer and issuer-view message hiding for the issuance surface; one-more unforgeability and full blindness are left open.

4 FROM BLIND SIGNATURES TO ARC

This section adapts the blind signature of Section 3 into an anonymous rate-limited credential (ARC) system. To do so, the holder signs the full message $\mu = \mathbf{m}\|\mathbf{k}$, where $\mathbf{m}$ is the attribute vector carried inside the credential and $\mathbf{k}$ is a uniformly sampled secret PRF key.

Rate-limiting via the secret key $\mathbf{k}$. Rate limiting is implemented by deterministic tags derived from a PRF keyed by the credential's secret $\mathbf{k}$. For a nonce $\text{nonce} \in \mathcal{D}$, the holder computes

$$\text{tag} = \mathbf{F}(\mathbf{k}, \text{nonce})$$

and attaches it as part of the presentation. It additionally proves correctness of tag with respect to $\mathbf{k}$ and that $\text{nonce} \in \mathcal{N}$, i.e. in the space of permitted nonces. Intuitively, if the holder never repeats a nonce, then tag is pseudorandom and yields unlinkable showings. If the holder repeats a nonce, the tag repeats deterministically, enabling the verifier to detect double-spending for that nonce. The verifier enforces its rate policy by maintaining state that records previously accepted $(\text{nonce}, \text{tag})$ values. Application context such as an epoch, service identifier, or session identifier can therefore be encoded into nonce , so context binding happens at showing time rather than by exposing extra public credential fields.

4.0.1 ARC protocol. We assume the same setup as in Section 3.0.2, and we write

$$\mathbf{A}_\mu = [\mathbf{A}_m \mid \mathbf{A}_k]$$

for the message/key split of the aligned commitment key. Moreover, define the relations

$$\mathcal{R}_{\text{Issue}, \text{ARC}} := \left\{ \left((\text{pp}_{\text{Com}}, \mathbf{c}, T, R_1, U, B), \begin{bmatrix} \mathbf{m} \\ \mathbf{k} \\ S \\ E \\ Z \end{bmatrix} \right) \mid \begin{array}{l} \text{pp}_{\text{Com}} = (\mathbf{A}_\mu, \mathbf{A}_s, \beta_{\text{msg}}) \\ \mathbf{c} = \mathbf{A}_m \mathbf{m} + \mathbf{A}_k \mathbf{k} \\ \mu = (\mathbf{m}\|\mathbf{k}) \\ \|\mathbf{m}, \mathbf{k}, S, E\|_\infty \leq \beta_{\text{msg}} \\ (B_3 - R_1) \odot Z = 1 \\ T = B_0 + U\mathbf{c} + Z \end{array} \right\}$$

Conditional policy soundness. Knowledge soundness of the post-sign proof yields a witness for the compiled showing relation of Section 5. Section 5.5 is the sole bridge from that compiled replay-space statement to the coefficient-domain statement used here. In particular, Corollary 5.4 yields a witness $(\mathbf{u}, T, M, K, R_0, R_1, Z)$ satisfying

$$(B_3 - R_1) \odot (T - B_0 - B_1(M + K) - B_2R_0) = 1$$

in R_q , together with

$$(B_3 - R_1) \odot Z = 1, \quad T = B_0 + B_1(M + K) + B_2R_0 + Z = h_{\text{tran}, M+K, (R_0, R_1)}(B)$$

and therefore

$$\mathbf{A}\mathbf{u} = h_{\text{tran}, M+K, (R_0, R_1)}(B).$$

Under that signature-side condition, together with one-more unforgeability of the blind issuance layer and correctness of the PRF, the signed key $\mathbf{k}$ determines the public tag uniquely for each nonce. Consequently, a verifier that records spent $(\text{nonce}, \text{tag})$ pairs and rejects same-nonce replays deterministically.

Conditional presentation unlinkability. Presentation unlinkability in Definition 2.23 is not implied by PRF security and post-sign zero knowledge alone. In that experiment the adversary controls the issuer during the two issuance sessions. Accordingly, unlinkability also depends on an issuance-layer privacy property, namely issuer-view hiding of the blind issuance surface. Under Proposition 3.2, post-sign zero knowledge hides the witness used during showing, and PRF pseudo-randomness makes the public tags computationally random across distinct fresh nonces. This yields the intended unlinkability claim for showing transcripts, subject to the explicit nonce-reuse caveat.

Attribute privacy and selective disclosure. In the core ARC relation of this paper, $\mathbf{m}$ remains hidden during showing. The message part $\mathbf{m}$ can encode attributes; the post-sign NIZK can be extended to prove predicates about $\mathbf{m}$ and to selectively reveal components, without changing the issuance protocol. We focus on the core rate-limiting construction and defer predicate engineering to the SmallWood programming model in Section 5.

5 SMALLWOOD PROGRAMMING MODEL

This section gives the SmallWood formulation of the ARC-SPRUCE NIZKs. The prover commits to packed witness rows, represented as low-degree row polynomials over a packing set Ω and masked on a disjoint set Ω' . The verifier then checks PACS-style constraints on those commitments. The main text records only the proof interface and the semantic issuance and showing statements. Concrete row layouts, gadget realizations, and proof details are deferred to Appendix C.

5.1 Proof interface

Fix a packing set $\Omega = \{\omega_1, \dots, \omega_s\} \subset \mathbb{F}_q$ and a disjoint hiding set $\Omega' = \{\omega'_1, \dots, \omega'_\ell\} \subset \mathbb{F}_q \setminus \Omega$. A compiled witness is

a matrix

$$W \in \mathbb{F}_q^{n \times s}.$$

For each row i , the prover interpolates a polynomial $P_i \in \mathbb{F}_q[X]$ of degree $< s + \ell$ such that $P_i(\omega_j) = W_{i,j}$ for all $\omega_j \in \Omega$, while $P_i(\omega'_h)$ is random on Ω' . The committed witness is therefore the family

$$P = (P_1, \dots, P_n),$$

committed through the DECS/LVCS/PCS stack.

SmallWood proves PACS constraint families on these committed rows. Parallel constraints are slotwise identities

$$f_a(P_1(\omega), \dots, P_n(\omega), \Theta_{a,1}(\omega), \dots, \Theta_{a,t_a}(\omega)) = 0 \quad \text{for every } \omega \in \Omega,$$

where the $\Theta_{a,j}$ are public helper rows. Aggregated constraints are domain-summed identities

$$\sum_{\omega \in \Omega} g_b(P_1(\omega), \dots, P_n(\omega), \Psi_{b,1}(\omega), \dots, \Psi_{b,u_b}(\omega)) = 0.$$

In ARC-SPRUCE, local algebra, replay bridges, and PRF checks are parallel; global carry or range checks are aggregated.

5.2 Issuance as a SmallWood statement

The public issuance statement is

$$(\text{pp}_{\text{Com}}, \mathbf{c}, T, R_1, U, B),$$

and the hidden witness is

$$(\mathbf{m}, \mathbf{k}, S, E, Z).$$

Because the target T is public, the compiled issuance proof is the simpler one. It commits rows encoding the message part, the hidden signed key, the linear-side randomness, and the inverse witness, together with auxiliary rows whose exact realization is deferred to the appendix.

Semantically, the proof establishes three claims. First, the public commitment $\mathbf{c}$ opens to the same hidden message-and-randomness witness used everywhere else. Second, the inverse witness satisfies

$$(B_3 - R_1) \odot Z = 1.$$

Third, the public target matches that same hidden witness through the aligned identity

$$T = B_0 + U\mathbf{c} + Z.$$

Thus the issuance proof certifies one coherent hidden opening witness for both $\mathbf{c}$ and the public target T .

5.3 Showing as a SmallWood statement

The public showing statement is the ARC verification context

$$(\text{pp}_{\text{Com}}, \text{nonce}, \text{tag}, \mathbf{A}, U, B, \beta_{\text{msg}}, \beta_{\text{sig}})$$

for the chosen nonce. The hidden witness consists of a signature witness $\mathbf{u}$, the hidden attribute vector, the hidden key $\mathbf{k}$, the hidden linear-side randomness R_0 , the hidden inverse-side randomness R_1 , and the auxiliary inverse witness Z .

The compiled showing proof commits three semantic layers. The first is a cert-side signature surface proving shortness of $\mathbf{u}$ and binding

$$T = \mathbf{A}\mathbf{u}.$$

The second is the source witness

$$T, M, K, R_0, R_1, Z,$$

where the source rows M and K carry the signed attribute and key lanes, while optional selector constraints can later be used for selective disclosure or predicate checks. The third is the replay witness: for each replay block b , the proof carries exact replay-image rows

$$\hat{T}_b, \hat{m}_b, \hat{R}_{0,b}, \hat{R}_{1,b}.$$

Writing $B_{i,b}^\ominus := \mathcal{F}_{\text{rep},b}(B_i)$, the proof enforces the exact source-to-replay bridges

$$\hat{T}_b = \mathcal{F}_{\text{rep},b}(T), \quad \hat{m}_b = \mathcal{F}_{\text{rep},b}(M + K),$$

$$\hat{R}_{0,b} = \mathcal{F}_{\text{rep},b}(R_0), \quad \hat{R}_{1,b} = \mathcal{F}_{\text{rep},b}(R_1),$$

and the blockwise replay residual

$$(B_{3,b}^\ominus - \hat{R}_{1,b}) \odot (\hat{T}_b - B_{0,b}^\ominus - B_{1,b}^\ominus \odot \hat{m}_b - B_{2,b}^\ominus \odot \hat{R}_{0,b}) = \mathcal{F}_{\text{rep},b}(1).$$

The adopted inverse witness is integrated directly in the source rows through

$$(B_3 - R_1) \odot Z = 1.$$

Hence the direct certified statement is a replay-space statement about one hidden witness: the same short signature yields T , the same source rows generate the replay-image rows, and the same hidden key is used by the PRF relation below.

5.4 PRF companion relation

The public tag is tied to the same hidden key carried by K through a companion PRF relation. In the main text only its semantic parts matter: selector constraints bind the PRF input lanes to the signed key, nonlinear checkpoint constraints certify the S-box steps of the permutation, and final feed-forward plus truncation bind the public pair (nonce, tag). The exact checkpoint schedule is an instantiation detail and is deferred to Appendix E and Appendix C.

5.5 Hiding and statement strength

The proof hides a family of row polynomials, not just a tuple of scalars. First, each row is masked by random values on Ω' , so unopened witness values are statistically hidden. Second, DECS/LVCS/PCS and the PACS batching expose only masked linear combinations and a few random-point openings outside Ω . Thus the verifier learns only the evaluations needed to check the constraints, and zero knowledge reduces to the hiding guarantees of the SmallWood commitment-opening stack and the simulator for the PACS layer.

The replay-space statement above is the only statement certified directly by the showing proof. The bridge back to the coefficient-domain relation used later is the following theorem suite.

Lemma 5.1 (Exact replay transform). *The public replay map*

$$\mathcal{F}_{\text{rep}} : R_q \rightarrow \mathcal{T}$$

is additive, multiplicative, and injective.

Lemma 5.2 (Replay-space certification implies the eliminated coefficient-domain BB-tran identity). *Assume the compiled showing relation of Section 5.3. Then*

$$(B_3 - R_1) \odot (T - B_0 - B_1(M + K) - B_2R_0) = 1$$

in R_q . Since the same relation also enforces $T = \mathbf{Au}$, it follows that

$$(B_3 - R_1) \odot (\mathbf{Au} - B_0 - B_1(M + K) - B_2R_0) = 1$$

in R_q .

Corollary 5.3 (Abstract rational-hash consequence). *Assume the compiled showing relation of Section 5.3. Then*

$$(B_3 - R_1) \odot Z = 1 \quad \text{and} \quad T = B_0 + B_1(M + K) + B_2R_0 + Z = h_{\text{tran}, M+K, (R_0, R_1)}$$

so in particular

$$\mathbf{Au} = h_{\text{tran}, M+K, (R_0, R_1)}(B).$$

Corollary 5.4 (Consequence of an accepting post-sign proof). *If the post-sign NIZK is knowledge sound for the compiled showing relation, then every accepting proof yields a witness $(\mathbf{u}, T, M, K, R_0, R_1, Z)$ satisfying the coefficient-domain BB-tran identity above and the abstract rational-hash relation.*

6 PARAMETERS AND INTEGRATION

This section explains how parameters interact across the ARC-SPRUCE stack: the lattice signature layer, the rational hash, the SmallWood proof system, and the PRF. Detailed parameter tables and extended benchmark data are deferred to Appendix D.

6.1 Parameter dependencies

All components share the same base ring R_q and modulus q :

- The signature map $A \in R_q^{n \times m}$ and hash key B are defined over R_q .
- SmallWood constraints are evaluated over the base field $\mathbb{F}_q$ and therefore use the same modulus q .
- The PRF is instantiated over $\mathbb{F}_q$ so that its constraints are native to the same field.

The coefficient bound β_{msg} governs the sampling ranges for the message and bounded randomness and the membership polynomials used by the NIZKs. For the aligned commitment and hiding theorem, the linear-side distributions χ_s, χ_e are instantiated as bounded-support or truncated distributions inside $[-\beta_{\text{msg}}, \beta_{\text{msg}}]$.

6.2 Sampler parameters

The trapdoor sampler must output short signatures u with overwhelming probability and with a distribution close to a discrete Gaussian over the appropriate lattice coset. This requires choosing:

- ring degree N and modulus q ,
- trapdoor quality,
- Gaussian widths for the sampler, and
- an acceptance predicate that yields a public verification bound β_{sig} .

We use an NTRU-style trapdoor together with a Gaussian preimage sampler; details and tuning rules are given in Appendix B.

6.3 vSIS and rational-hash parameters

Rational-hash admissibility. The BB-tran target is evaluated only when $B_3 - R_1 \in R_q^\times$. In the issuance model, this admissibility condition is enforced during honest target derivation by restart. The compiled pre-sign proof of Section 5.2 certifies the aligned public identity

$$T = B_0 + U\mathbf{c} + Z, \quad (B_3 - R_1) \odot Z = 1,$$

while the compiled post-sign proof certifies the eliminated coefficient-domain relation

$$(B_3 - R_1) \odot (T - L_B(m, R_0)) = 1.$$

No wrap-around in bounds. Membership checks for $[-\beta_{\text{msg}}, \beta_{\text{msg}}]$ are implemented via vanishing polynomials over $\mathbb{F}_q$, so one requires $\beta_{\text{msg}} \ll q$ to avoid modular wrap-around artifacts.

Non-blind reduction parameters. The non-blind reduction imported from DKLW25 is parameterized by a norm bound β_{DKLW} in the norm used in [DKLW25a], by the inadmissibility bound δ , and by the predicate condition P . For the present instantiation, the external reduction basis is the BB-tran / GenISIS_f / IntGenISIS_f line summarized in Lemma 2.14, which places the non-blind signature layer on the corresponding sEUF-CMA footing. The public verifier bound β_{sig} used by ARC-SPRUCE should therefore not be identified with β_{DKLW} without a separate norm-comparison lemma. In this section, β_{sig} is recorded only as the public acceptance bound of the concrete verifier.

6.4 SmallWood parameters

SmallWood introduces parameters controlling packing, masking, and soundness:

- the packing size $s = |\Omega|$,
- the tail length ℓ for HVZK masking and degree enforcement,
- the spot-check count ℓ' and the number of masks ρ for PIOP soundness,
- the degree bounds for witness rows and constraint-derived masks, and
- the small-field knob θ when using an extension-field variant.

The pre-sign proof is dominated by carrier membership, decode bridges, the inverse-witness check, and the aligned public-target identity $T = B_0 + U\mathbf{c} + Z$. The post-sign proof adds the signature shortness gadget, the full replay-image showing relation in eliminated BB-tran form, and the PRF checkpoint constraints.

Full replay-image geometry. Let $n_{\text{cols}}^{\text{rep}}$ denote the row width used for the replay blocks, and write

$$B_{\text{rep}} := \left\lceil \frac{N}{n_{\text{cols}}^{\text{rep}}} \right\rceil.$$

In the full replay-image showing model, each replayed ring object is represented by B_{rep} replay blocks. Accordingly, the post-sign witness carries

$$4B_{\text{rep}}$$

replay rows for the full images of Au , $M + K$, R_0 , and R_1 , in addition to the cert-side signature surface, source rows, and PRF companion rows.

Replay width as a proof-size knob. Increasing $n_{\text{cols}}^{\text{rep}}$ decreases the replay block count B_{rep} and hence the number of replay-image rows and associated bridge families. This tends to reduce transcript size. The tradeoff is that larger row width increases the degree of the committed row polynomials and therefore affects the LVCS/PCS soundness and opening costs. The sizing discussion should therefore treat $n_{\text{cols}}^{\text{rep}}$ as the main knob controlling the transcript-size tradeoff of the full replay-image showing proof.

Replay-transform obligations for the showing theorem. The coefficient-domain consequence theorem of Section 5.5 relies on the exact full replay transform $\mathcal{F}_{\text{rep}}$ and on the exact bridge equalities for the replay-image rows. In the full replay-image model, injectivity is supplied by the transform itself rather than by an additional zero-detection assumption on a reduced replay map; the replay residual transports the eliminated BB-tran relation directly.

6.5 PRF parameters

The PRF must be secure at the target security level, efficiently representable in constraints, and compatible with the base field $\mathbb{F}_q$. We use a Poseidon2-like permutation with feed-forward and truncation as a concrete instantiation. Its parameters include the state width t , round counts (R_F, R_P) , MDS matrices, round constants, S-box exponent d , and truncation length ℓ_{tag} . We require $\ell_{\text{tag}} \geq \lceil \lambda / \log_2(q) \rceil$ so the tag contains λ bits of entropy. The algorithm is given in Section 4.1, while the checkpointed constraint shape used in the post-sign proof is stated in Section 5.4. Appendix E records extended notes.

6.6 Sizing discussion

Concrete proof sizes depend jointly on the sampler parameters, the SmallWood masking parameters, the replay-block width, and the PRF checkpoint schedule. In the full replay-image showing model, enlarging the replay-block width decreases the number of committed replay rows, while the degree and opening costs of each row increase. Appendix D records representative parameter tuples and measurement data; the present section records only the qualitative coupling between those choices.

7 IMPLEMENTATION RESULTS

We implemented ARC-SPRUCE in Go. Ring arithmetic and NTTs are provided by Lattigo v4.1.1, while the credential and commitment logic, the retained SmallWood proof stack (PIOP together with DECS/LVCS), and the Poseidon2-like

Metric	Issuance	Showing
Arithmetic / relation	Shared R_q with $N = 1024$, $q = 1054721$, <code>bb.tran</code> ; public-target pre-sign proof	Same ring and relation; retained reduced-replay post-sign proof
PRF / shortness	Credential fixes a Poseidon2-like PRF with $t = 20$, $d = 3$, $R_F = 8$, $R_P = 19$, and key/nonce/tag lengths 8/12/7; no shortness subproof	Same PRF instance with <code>output.audit</code> , group rounds = 2, check-point samples = 8; nested hidden shortness proof (<code>SigShortnessV6</code>), outer profile <code>r12285.11_research</code>
SmallWood tuple	(16, 18, 2, 2, 1, 19, 96, 4096), $d_Q = 312$	(16, 18, 3, 2, 3, 26, 32, 4096), $\kappa = \{0, 11, 0, 11\}$, $d_Q = 312$
Witness geometry	22 witness rows, 2 mask rows	22 logical rows, 19 committed rows, 60 mask rows
Paper transcript	15,915 B	26,587 B
Verifier payload	49,153 B	47,848 B
Prover time	0.194 s	2.854 s
Verifier time	n/a	86.1 ms
Dominant terms	Payload: RowOpening 17,399, QR 15,621, QOpening 15,063. Paper: Auth 4,998, R 4,562, Pdecs 2,269	Paper: SigShortness 9,445, Pdecs 6,627, Q 4,622, Auth 2,358, R 2,081

Table 1: Implementation measurements for the measured issuance instance and the retained reduced-replay showing instance. SmallWood tuples are listed in the order $(NCols, \ell, \ell', \rho, \theta, \eta, LVCSNCols, NLeaves)$. “Paper transcript” denotes the paper-faithful transcript accounting, whereas “verifier payload” denotes the current serialized proof object consumed by the implementation.

PRF are implemented in custom Go code over the same field $\mathbb{F}_q$. Table 1 reports the single measured issuance instance and the retained reduced-replay showing instance used by the current implementation. Benchmarks were obtained on a Linux `x86_64` host with Go 1.26.0 on an Intel i7-10875H.

For the measured parameter sets, issuance is smaller because it only certifies the public-target pre-sign relation. Showing additionally carries the reduced replay surface, the PRF companion constraints, and a nested hidden shortness proof, so its size is dominated by the shortness bucket together with row-opening and quotient-related Q material. Beyond the shortness proof, the remaining replay burden is concentrated in the PRF checkpoint and final-tag rows and in the two source-product rows induced by the cleared `bb.tran` relation. Conservatively, the retained reduced showing instance stays below 26 KB in paper-transcript size and verifies in under 0.1 s, while issuance proving stays below 0.2 s for the measured instance.

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A ADDITIONAL PRELIMINARIES

A.1 Security assumptions for Lattices

Fix a public left factor F , hint space H , target space G , predicate P , and query bound Q .

- **Hint-vSIS.** The adversary chooses a query set $\mathcal{Q} \subseteq H$ and a fresh target function $g^* \in G \setminus \mathcal{Q}$ before seeing the sampled matrix A . It then receives short preimages of the queried hints $q(A)$ under $F(A)$ for every $q \in \mathcal{Q}$, and must output a non-zero short vector u^* such that $F(A)u^* = g^*(A)$. The instance is restricted by the predicate condition $P(F \cup \mathcal{Q} \cup (\{g^*\} \setminus \{0\})) = 1$.
- **s-Hint-vSIS.** This is the strong variant of the same experiment. The adversary still chooses the hint set $\mathcal{Q}$ before seeing A , but it is allowed to choose the fresh target g^* only after seeing A and the hinted preimages.
- **s- $\mathbb{S}$ -Hint-vSIS.** This is the strong random-hinted variant. It is the same as **s-Hint-vSIS**, except that the query set $\mathcal{Q}$ is sampled uniformly at random from H rather than chosen by the adversary.

Now fix a keyed evaluation family $f : K \times M \times X \rightarrow R_q^n$.

- **GenISIS $_f$.** The challenger samples a public key $\kappa \leftarrow K$ and a uniform matrix $A \xleftarrow{\$} R_q^{n \times m}$. It provides a preimage oracle that, on input $\mu \in M$, samples $\chi \leftarrow X$, sets $t = f(\kappa, \mu, \chi)$, returns a short preimage s such that $As = t$, and records the tuple (s, μ, χ) . The adversary wins if it outputs a fresh tuple (s^*, μ^*, χ^*) not previously returned by the oracle such that $As^* = f(\kappa, \mu^*, \chi^*)$ and $0 < \|s^*\| \leq \beta$.
- **IntGenISIS $_f$.** This is an interactive version where the challenger offers a signing oracle with an additional linear masking term.

[DKLW25b] shows how this interactive problem reduces back to GenISIS $_f$ under increased parameters.

A.2 Centering map and carries

Let $\text{BoundB} \in \mathbb{Z}_{>0}$ and $\Delta = 2\text{BoundB} + 1$. Define the centering map $\text{center} : [-2\text{BoundB}, 2\text{BoundB}] \rightarrow [-\text{BoundB}, \text{BoundB}]$ by

$$\text{center}(x) = \begin{cases} x & \text{if } x \in [-\text{BoundB}, \text{BoundB}], \\ x + \Delta & \text{if } x \in [-2\text{BoundB}, -\text{BoundB} - 1], \\ x - \Delta & \text{if } x \in [\text{BoundB} + 1, 2\text{BoundB}]. \end{cases}$$

Applied coefficient-wise, this map allows unbiased combination of two bounded values when at least one is uniform.

Lemma A.1 (Centering preserves uniformity). *Fix any $a \in [-\text{BoundB}, \text{BoundB}]$. If $b \xleftarrow{\$} [-\text{BoundB}, \text{BoundB}]$ then $\text{center}(a + b)$ is uniform over $[-\text{BoundB}, \text{BoundB}]$.*

A.3 Vanishing polynomials for membership constraints

Membership constraints are expressed using vanishing polynomials over $\mathbb{F}_q$ with signed lifting.

Definition A.2 (Vanishing polynomial for $[-\text{BoundB}, \text{BoundB}]$). Let $\text{BoundB} \in \mathbb{Z}_{>0}$. Define

$$P_{\text{BoundB}}(X) = \prod_{i=-\text{BoundB}}^{\text{BoundB}} (X - \langle i \rangle_q) \in \mathbb{F}_q[X].$$

For $x \in \mathbb{F}_q$, we have $x \in [-\text{BoundB}, \text{BoundB}]$ (via signed lifting) iff $P_{\text{BoundB}}(x) = 0$ in $\mathbb{F}_q$.

Definition A.3 (Vanishing polynomial for $\{-1, 0, 1\}$). Define $P_1(X) = (X + 1)X(X - 1) \in \mathbb{F}_q[X]$. Then $P_1(x) = 0$ iff $x \in \{-1, 0, 1\}$.

B GAUSSIAN PREIMAGE SAMPLING

This appendix details the Gaussian preimage sampler used by the issuer to sign a target $t \in R_q^n$ by producing a short preimage $u \in R^m$ such that $Au = t$ in R_q^n . Our instantiation follows the vSIS-BB-tran setting [DKLW25a] and relies on an NTRU trapdoor generated via Antrag together with a hybrid two-plane sampler [ENS⁺23]. In the concrete ARC-SPRUCE instantiation, the signed target is the single-coordinate h_{tran} value produced by the aligned issuance layer.

B.1 The preimage sampling problem

Given a public linear map $A \in R_q^{n \times m}$ and a target $t \in R_q^n$, the signer samples

$$u \leftarrow \text{SampPre}(\text{td}_A, t) \quad \text{such that} \quad Au \equiv t \pmod{q}$$

and u is short (according to a public bound β_{sig}). Equivalently, u lies in the coset lattice

$$\Lambda_{\perp}^t(A) := \{x \in R^m : Ax \equiv t \pmod{q}\}.$$

We require: (i) correctness ($Au = t$ and the norm test fails only with negligible probability), (ii) (approx.) Gaussianity of u over the coset (statistical hiding of the trapdoor), and (iii) a tight tail bound that supports efficient ZK range/norm checks in the post-sign proof.

B.2 NTRU trapdoor form and quality

We work with the rank-2 NTRU module over the cyclotomic ring $R = \mathbb{Z}[X]/(X^N + 1)$ and its quotient R_q . An NTRU public key has the form $h = gf^{-1} \pmod{q}$ for secret $f, g \in R$. The corresponding module is

$$\mathbf{L}_{\text{NTRU}}(h) = \{(u, v) \in R^2 : uh - v \equiv 0 \pmod{q}\}.$$

A trapdoor basis of $\mathbf{L}_{\text{NTRU}}(h)$ is

$$B_{f,g} = \begin{pmatrix} f & F \\ g & G \end{pmatrix}, \quad \text{with} \quad fG - gF = q,$$

whose columns are short and satisfy $g/f \equiv G/F \equiv h \pmod{q}$. Following [ENS⁺23], we measure trapdoor quality via per-slot energies under the canonical embeddings φ_i , requiring a uniform window around q . Concretely, for a quality parameter $\alpha > 1$ we target

$$\frac{q}{\alpha^2} \leq |\varphi_i(f)|^2 + |\varphi_i(g)|^2 \leq \alpha^2 q \quad \text{for all slots } i.$$

Smaller α corresponds to tighter trapdoors and, for fixed N, q , to shorter signatures for the same security.

B.3 Sampler definition and tuning

Our sampler is a ring-based hybrid two-plane sampler in the evaluation domain. The sampler's main tuning parameters are:

- modulus q and ring degree N ,
- trapdoor quality α ,
- acceptance radius R_{acc} (controls rejections and ℓ_∞/ℓ_2 tails),
- per-slot widths $\sigma_k[i]$ (two planes $k \in \{1, 2\}$).

Per-slot calibration. Let $\tilde{b}_k[i]$ denote the per-slot Gram-Schmidt lengths for plane k at slot i . A standard calibration equalizes acceptance probability across slots:

$$\sigma_k[i] = \sqrt{\frac{\alpha^2 q R_{\text{acc}}^2}{\|\tilde{b}_k[i]\|^2} - R_{\text{acc}}^2} \quad (k \in \{1, 2\}).$$

This prevents weak slots and yields consistent output variance.

Hiding threshold (smoothing). To achieve statistical hiding, we enforce a floor $\sigma_{\min}$ such that $\sigma_k[i] \geq \sigma_{\min}$ for all k, i , where $\sigma_{\min}$ exceeds the smoothing parameter of the relevant lattice cosets. In practice, we apply a global scale factor to raise all $\sigma_k[i]$ above the theoretical floor with slack.

B.4 Annulus sampling (Antrag key generation)

Antrag replaces trial-and-error trapdoor generation with direct annulus sampling in the Fourier domain [ENS⁺23]. For each conjugate pair (there are $N/2$ independent slots), the key generator samples $(|\varphi_i(f)|, |\varphi_i(g)|)$ uniformly in an annulus arc $A^+(r, R)$ with

$$\sqrt{q}/\alpha < r < R < \alpha\sqrt{q},$$

chooses random phases, inverse-embeds to the coefficient domain, and rounds to obtain $(f, g) \in R^2$. By selecting a “middle” annulus window (see [ENS⁺23]) one ensures that the rounded trapdoor satisfies the desired α -window with high probability and with modest repetition.

B.5 Hybrid two-plane sampler (details)

We sketch the two-plane hybrid sampler in the notation of [ENS⁺23]. Let td_A encode an NTRU trapdoor for A and let $t \in R_q^n$ be the target. Write $\langle \cdot \rangle_q$ for signed coefficient lift and define center_q as coefficient-wise centering into $(-\frac{q}{2}, \frac{q}{2}]$.

Algorithm 2 $\text{SampPre}(\text{td}_A, t)$: two-plane hybrid sampler (sketch)

- 1: Lift t to coefficients and set $c \leftarrow \text{center}_q(t) \in (-\frac{q}{2}, \frac{q}{2}]^n$.
 - 2: Using td_A , compute per-slot widths $\sigma_k[i]$ (two planes) calibrated to trapdoor quality.
 - 3: **repeat**
 - 4: Sample slot-wise Gaussian pairs on the two planes, reconstruct a coefficient-domain candidate u .
 - 5: Accept iff the norm predicate holds for u (e.g., ℓ_2 and/or ℓ_∞ bound).
 - 6: **until** accepted
 - 7: **return** u .
-

Public relations. Downstream verification and proofs use only the public relation $Au = t$ and shortness of u (bounded by β_{sig}), checked by the holder on receipt and enforced by the issuer's rejection sampling. The post-sign NIZK then proves knowledge of such a short u without revealing it.

C SMALLWOOD PROOF-STACK DETAILS

This appendix records the proof-stack geometry for the compiled relations of Section 5. It specifies committed witness-row families, public lifts, PACS splits, and the PCS/LVCS/DECS opening structure used by the prover and verifier. It does not strengthen the statement layer beyond Section 5; in particular, coefficient-domain consequences are obtained exactly by the theorem suite stated there.

C.1 Deferred row inventories and bridge formulas

Section 5 groups the witness rows semantically. We record here the exact compiled surfaces and bridge formulas used by the issuance and showing relations.

Issuance witness surface. The committed issuance witness consists of carrier rows for the signed message lanes and the linear-side randomness, together with decoded source rows

$$M, K, S, E, Z.$$

The exact carrier-to-source bridges decode the committed message/key lanes and the linear-side randomness lanes into these source rows. The aligned commitment is then enforced directly by

$$\mathbf{c} = \mathbf{A}_m M + \mathbf{A}_k K + \mathbf{A}_s S + E,$$

and the inverse witness is enforced by

$$(B_3 - R_1) \odot Z = 1.$$

Finally, the public issuance target is tied to the same committed witness by the aligned identity

$$T = B_0 + U\mathbf{c} + Z.$$

Thus the pre-sign statement binds one coherent hidden witness to the commitment, the inverse witness, and the public target.

Showing witness surface. The committed showing witness consists of:

- (1) the signature limb rows $L_{c,b,\ell}$;
- (2) the carrier rows C^M and C^{ctr} ;
- (3) the source rows

$$T, M, K, R_0, R_1, Z;$$

- (4) the replay-image block families

$$\hat{T}_0, \dots, \hat{T}_{B_{\text{rep}}-1}, \quad \hat{m}_0, \dots, \hat{m}_{B_{\text{rep}}-1},$$

$$\hat{R}_{0,0}, \dots, \hat{R}_{0,B_{\text{rep}}-1}, \quad \hat{R}_{1,0}, \dots, \hat{R}_{1,B_{\text{rep}}-1};$$

- (5) the PRF companion rows.

The cert-side signature witness u is reconstructed from the limb surface by the public digit-collapse map of the shortness gadget and is tied to the source row T by the source-side equation $T = \mathbf{A}u$. The carrier-to-source bridges are

$$M = \text{Dec}_M^{(1)}(C^M), \quad K = \text{Dec}_M^{(2)}(C^M),$$

$$R_0 = \text{Dec}_{\text{ctr}}^{(1)}(C^{\text{ctr}}), \quad R_1 = \text{Dec}_{\text{ctr}}^{(2)}(C^{\text{ctr}}).$$

The source-side inverse witness satisfies

$$(B_3 - R_1) \odot Z = 1.$$

For every replay block $b \in \{0, \dots, B_{\text{rep}} - 1\}$, the exact source-to-replay bridges are

$$\hat{T}_b = \mathcal{F}_{\text{rep},b}(T), \quad \hat{m}_b = \mathcal{F}_{\text{rep},b}(M + K),$$

$$\hat{R}_{0,b} = \mathcal{F}_{\text{rep},b}(R_0), \quad \hat{R}_{1,b} = \mathcal{F}_{\text{rep},b}(R_1).$$

C.2 Domains and packing

Fix a base field $\mathbb{F}_q$, an evaluation set $\Omega \subset \mathbb{F}_q$ of size s , and a disjoint tail set $\Omega' \subset \mathbb{F}_q \setminus \Omega$ of size ℓ . A witness row is interpolated into a polynomial of degree $\leq s + \ell - 1$ whose values on Ω are the packed coordinates and whose values on Ω' are random masks. The prover commits to these explicit-domain row polynomials, and the verifier later replays the active constraint residuals from openings of the same polynomials.

C.3 DECS (Degree-Enforcing Commitment Scheme)

DECS commits to evaluations of witness polynomials and enforces that they arise from low-degree polynomials rather than arbitrary vectors. In our instantiation, commitments are implemented by a Merkle tree over an NTT-sized domain, while degree soundness is obtained by checking masked low-degree relations at verifier-chosen indices.

Algorithm 3 DECS.DeriveGamma(root, η, r, q) $\rightarrow \Gamma$ (verifier and prover)

```

1: Initialize counter  $\text{ctr} \leftarrow 0$ ; let  $L \leftarrow \lfloor \frac{2^{64}}{q} \rfloor \cdot q$ .
2: for  $k = 0$  to  $\eta - 1$  do
3:   for  $j = 0$  to  $r - 1$  do
4:     repeat
5:        $x \leftarrow \text{SHA256}(\text{root} \parallel \text{LE64}(\text{ctr}))$  interpreted as
        a 64-bit integer
6:        $\text{ctr} \leftarrow \text{ctr} + 1$ 
7:     until  $x < L$ 
8:      $\Gamma_{k,j} \leftarrow x \bmod q$ 
9: return  $\Gamma$ 

```

Algorithm 4 DECS.VerifyEval($\text{root}, \Gamma, R, \text{open}$) (verifier)

Require: Domain size N , modulus q , mask count η , committed row count r .

```

1: for each opened index  $e$  do
2:   Verify Merkle authentication for  $e$  against  $\text{root}$ ; reject
   on failure.
3:   for each mask row  $k < \eta$  do
4:     Check  $R_k^*[e] \stackrel{?}{=} M_k(e) + \sum_{j=0}^{r-1} \Gamma_{k,j} P_j(e) \pmod{q}$ .
5: return accept

```

C.4 LVCS (Linear Vector Commitments)

LVCS lifts DECS into a commitment to packed witness rows that supports succinct openings of $\mathbb{F}_q$ -linear forms of those rows. SmallWood uses LVCS both as a batching layer and as the core of its hash-based PCS: the verifier requests openings of linear forms corresponding to replay residuals, and DECS ensures that the opened values come from low-degree committed row polynomials.

Algorithm 5 LVCS.CommitNitWithParams($\text{ringQ}, \{r_j\}_{j < r}, \ell, \text{params}$) $\rightarrow (\text{root}, \text{ProverKey})$ (prover)

Require: Ring R_q of size N , packed rows $r_j \in \mathbb{F}_q^{n_{\text{cols}}}$, tail length $\ell > 0$.

```

1: Sample mask  $j \xleftarrow{\$} \mathbb{F}_q^\ell$  for each row  $j$  (HVZK tails).
2: Interpolate each row to  $P_j(X) \leftarrow \text{interpolateRow}(\text{ringQ}, r_j, \text{mask}_j, n_{\text{cols}}, \ell)$ .
3: Commit to  $\{P_j\}$  via DECS and obtain Merkle root  $\text{root}$ .
4: Derive  $\Gamma \leftarrow \text{DECS.DeriveGamma}(\text{root}, \eta, r, q)$  and cache
   values needed for openings.
5: return ( $\text{root}, \text{ProverKey}$ ).

```

Algorithm 6 LVCS.CommitStep1(root) $\rightarrow \Gamma$ (verifier)

```

1: Store  $\text{root}$  and derive  $\Gamma \leftarrow \text{DECS.DeriveGamma}(\text{root}, \eta, r, q)$ .
2: return  $\Gamma$ .

```

Algorithm 7 LVCS.CommitStep2($\{R_k\}_{k<\eta} \rightarrow \text{accept/reject}$ (verifier))

- 1: **for** each mask row R_k **do**
 - 2: Reject if $\deg(R_k) > d$ (degree guard before openings).
 - 3: **return** accept.
-

Algorithm 8 LVCS.EvalInitMany(ringQ, ProverKey, $\{\text{Coeffs}_k\}_{k<m} \rightarrow \{\bar{v}_k\}_{k<m}$ (prover))

Require: Linear forms $\text{Coeffs}_k \in \mathbb{F}_q^r$ for $k = 0, \dots, m-1$.

- 1: Let $\ell \leftarrow |\text{mask}_j|$.
 - 2: **for** $k = 0$ to $m-1$ **do**
 - 3: Initialize $\bar{v}_k \leftarrow 0 \in \mathbb{F}_q^\ell$.
 - 4: Set $\bar{v}_k \leftarrow \sum_{j=0}^{r-1} \text{Coeffs}_k[j] \cdot \text{mask}_j \in \mathbb{F}_q^\ell$.
 - 5: **return** $\{\bar{v}_k\}_{k<m}$.
-

Algorithm 9 LVCS.ChooseE($\ell, n_{\text{cols}} \rightarrow E$ (verifier))

Require: Ring size N ; packed domain size $n_{\text{cols}} = |\Omega|$; masked slab indices $[n_{\text{cols}}, n_{\text{cols}} + \ell)$.

- 1: Sample $E \subset \{n_{\text{cols}} + \ell, \dots, N-1\}$ uniformly without replacement with $|E| = \ell$.
 - 2: **return** E .
-

Algorithm 10 LVCS.EvalFinish(ProverKey, $E \rightarrow \text{Opening}$ (prover))

- 1: $\text{open} \leftarrow \text{ProverKey.DecsProver.EvalOpen}(E)$.
 - 2: **return** $\text{Opening}\{\text{DECSOpen} = \text{open}\}$.
-

Algorithm 11 LVCS.EvalStep2($\{\bar{v}_k\}, E, \text{open}, C, v_{\text{targets}}$ (verifier))

Require: Coeffs $C \in \mathbb{F}_q^{m \times r}$, public prefixes $v_{\text{targets}} \in (\mathbb{F}_q^{n_{\text{cols}}})^m$.

- 1: Split open into mask openings for $[n_{\text{cols}}, n_{\text{cols}} + \ell)$ and tail openings at E .
 - 2: Verify the DECS openings (degree soundness and Merkle authentication); reject on failure.
 - 3: **for** each $k < m$ and each masked index $i < \ell$ **do**
 - 4: Check $\sum_{j=0}^{r-1} C[k, j] \cdot \text{maskOpen.Pvals}[i][j] \stackrel{?}{=} \bar{v}_k[i] \pmod{q}$.
 - 5: Reconstruct each batched polynomial $Q_k(X)$ from its Ω -prefix $v_{\text{targets}}[k]$ and mask tail $\bar{v}_k$.
 - 6: Check tail consistency of Q_k against the opened linear combinations at points in E .
 - 7: **return** accept.
-

C.5 LVCS and PCS/PIOP

LVCS provides the linear opening queries needed by the polynomial oracle model, while DECS provides degree soundness for the committed rows. SmallWood combines the two into a transparent hash-based PCS used by the PACS and PIOP layers.

C.6 What DECS/LVCS/PCS bind in ARC-SPRUCE

At a high level, the SmallWood stack binds *all committed witness rows of one statement* under a single PCS root, and it enforces that every opened value is consistent with one low-degree polynomial view of those rows. In our ARC instantiation this binding matters in three ways:

- **Row binding and low degree (DECS).** DECS ensures that every committed row is the low-degree polynomial matching the packed values on Ω and the random tails on Ω' .
- **Replay binding across subsystems (LVCS).** The verifier reuses the same opened row values to replay every active constraint family. In issuance this covers carrier membership, decode bridges, the aligned commitment, the inverse-witness check, and the direct public-target identity $T = B_0 + U\mathbf{c} + Z$. In showing it covers signature shortness, the signature-side bridges to the full replay-image family $\hat{T}_0, \dots, \hat{T}_{B_{\text{rep}}-1}$, the non-sign bridges to $\hat{m}_0, \dots, \hat{m}_{B_{\text{rep}}-1}$, $\hat{R}_{0,0}, \dots, \hat{R}_{0,B_{\text{rep}}-1}$, and $\hat{R}_{1,0}, \dots, \hat{R}_{1,B_{\text{rep}}-1}$, the blockwise replay residuals, and the PRF companion constraints.
- **Single committed witness.** Because all committed rows live under one PCS root, the prover cannot use one hidden message for the hash relation and another for the PRF or the commitment. All replay checks refer to one coherent witness assignment.

C.7 Openings requested in issuance and showing

Both proofs follow the same pattern: the prover commits to all statement rows as explicit-domain polynomials, the verifier derives Fiat-Shamir challenges, and the prover opens the committed rows at verifier-chosen points. The verifier then recomputes the active residuals from those openings and checks the parallel replay identities at E' together with the aggregated Ω -sum identities.

Pre-sign witness surface and replay. The committed pre-sign rows decode to the source rows

$$M, K, S, E, Z.$$

The verifier replays the carrier membership relations, decode bridges, the aligned commitment equation

$$\mathbf{c} = \mathbf{A}_m M + \mathbf{A}_k K + \mathbf{A}_s S + E,$$

the inverse-witness equation

$$(B_3 - R_1) \odot Z = 1,$$

and the direct aligned target identity

$$T = B_0 + U\mathbf{c} + Z.$$

This appendix therefore records the compiled relation $R_{\text{Issue}}^{\text{cmp}}$ only.

Post-sign witness surface and replay. The committed post-sign rows consist of:

- (1) the signature limb rows;
- (2) the carrier rows C^M and C^{ctr} ;
- (3) the decoded non-sign source rows M, K, R_0, R_1 ;
- (4) the full replay-image block families

$$\begin{aligned} \hat{T}_0, \dots, \hat{T}_{B_{\text{rep}}-1}, \quad \hat{m}_0, \dots, \hat{m}_{B_{\text{rep}}-1}, \\ \hat{R}_{0,0}, \dots, \hat{R}_{0,B_{\text{rep}}-1}, \quad \hat{R}_{1,0}, \dots, \hat{R}_{1,B_{\text{rep}}-1}; \end{aligned}$$

- (5) the PRF companion rows.

The verifier replays:

- (1) the shortness families on the cert-side signature surface;
- (2) the carrier membership and decode families;
- (3) the signature-side bridges

$$\hat{T}_b = \mathcal{F}_{\text{rep},b}(\mathbf{A}u)$$

for every block b ;

- (4) the non-sign transform bridges

$$\hat{m}_b = \mathcal{F}_{\text{rep},b}(M+K), \quad \hat{R}_{0,b} = \mathcal{F}_{\text{rep},b}(R_0), \quad \hat{R}_{1,b} = \mathcal{F}_{\text{rep},b}(R_1)$$

for every block b ;

- (5) the blockwise replay residual

$$(B_{3,b}^{\ominus} - \hat{R}_{1,b}) \odot (\hat{T}_b - B_{0,b}^{\ominus} - B_{1,b}^{\ominus} \odot \hat{m}_b - B_{2,b}^{\ominus} \odot \hat{R}_{0,b}) = \mathcal{F}_{\text{rep},b}(1)$$

for every block b ; and

- (6) the PRF checkpoint constraints.

The theorem stated in Section 5.5 then transports this full replay-image statement to the coefficient-domain BB-tran identity.

Replay-transform realization. The replay transform

$$\mathcal{F}_{\text{rep}} : R_q \rightarrow \mathcal{T}$$

is the exact public transform used by the replay basis, partitioned into the replay blocks

$$\mathcal{F}_{\text{rep},0}, \dots, \mathcal{F}_{\text{rep},B_{\text{rep}}-1}.$$

The bridge families above certify that the replay-image rows are exactly the block projections of $\mathcal{F}_{\text{rep}}(\mathbf{A}u)$, $\mathcal{F}_{\text{rep}}(M+K)$, $\mathcal{F}_{\text{rep}}(R_0)$, and $\mathcal{F}_{\text{rep}}(R_1)$. These are precisely the identities used in Lemma 5.2.

Proof of Lemma 5.1. The replay basis is chosen so that

$$\mathcal{F}_{\text{rep}} : R_q \longrightarrow \mathcal{T}$$

is the corresponding exact CRT/NTT transform of R_q . Its additivity and multiplicativity are the defining algebraic properties of that transform, and injectivity follows because it is an isomorphism onto its image.

Proof of Lemma 5.2. For every replay block b , the source-to-replay bridges and the public lift definition give

$$\hat{T}_b = \mathcal{F}_{\text{rep},b}(T), \quad \hat{m}_b = \mathcal{F}_{\text{rep},b}(M+K),$$

$$\hat{R}_{0,b} = \mathcal{F}_{\text{rep},b}(R_0), \quad \hat{R}_{1,b} = \mathcal{F}_{\text{rep},b}(R_1),$$

$$B_{i,b}^{\ominus} = \mathcal{F}_{\text{rep},b}(B_i) \quad \text{for } i \in \{0, 1, 2, 3\}.$$

Substituting these identities into the blockwise replay residual yields

$$\mathcal{F}_{\text{rep},b}(B_3 - R_1) \odot (\mathcal{F}_{\text{rep},b}(T) - \mathcal{F}_{\text{rep},b}(B_0) - \mathcal{F}_{\text{rep},b}(B_1(M+K)) - \mathcal{F}_{\text{rep},b}(B_2R_0))$$

for every b . By additivity and multiplicativity of $\mathcal{F}_{\text{rep},b}$, this is

$$\mathcal{F}_{\text{rep},b}((B_3 - R_1) \odot (T - B_0 - B_1(M+K) - B_2R_0)) = \mathcal{F}_{\text{rep},b}(1)$$

for every b . Concatenating the replay blocks gives

$$\mathcal{F}_{\text{rep}}((B_3 - R_1) \odot (T - B_0 - B_1(M+K) - B_2R_0)) = \mathcal{F}_{\text{rep}}(1).$$

By Lemma 5.1, $\mathcal{F}_{\text{rep}}$ is injective. Hence

$$(B_3 - R_1) \odot (T - B_0 - B_1(M+K) - B_2R_0) = 1$$

in R_q . Since the compiled showing relation also enforces $T = \mathbf{A}u$, this immediately implies

$$(B_3 - R_1) \odot (\mathbf{A}u - B_0 - B_1(M+K) - B_2R_0) = 1$$

in R_q .

The Ω -sum check. The Ω -sum check is active in both proofs. In issuance it aggregates the carrier/decode checks together with the aligned commitment and public-target identity. In showing it enforces the full signature-side bridge family together with the full non-sign transform bridges. Thus the Σ_{Ω} check is part of the statement layer in both issuance and showing.

C.8 PACS/PIOP interface

SmallWood's PACS/PIOP layer checks that constraint polynomials vanish on Ω and at random points outside Ω , by committing to witness rows and mask polynomials and then opening them at verifier-chosen coordinates. The following algorithms summarize the high-level structure used in our proofs.

Algorithm 12 PACS.PIOPCommitOracle_{Prover}(params) $\rightarrow$ pcsCom

- 1: Interpolate witness rows into $P_1, \dots, P_n \in \mathbb{F}_q[X]_{\leq s+\ell'-1}$ with ℓ' random blind evaluations outside Ω .
 - 2: Sample mask polynomials $M_1, \dots, M_{\rho} \in \mathbb{F}_q[X]_{\leq d_Q}$ such that $\sum_{\omega \in \Omega} M_i(\omega) = 0$ for all i .
 - 3: Commit to $P \parallel M$ via PCS (implemented through LVCS/DECS); obtain pcsCom.
 - 4: **return** pcsCom.
-

Algorithm 13 PACS.PIOPBatchCombine_{Prover}(pcsCom, Γ') $\rightarrow$ Q

1: Using P, M , form batched polynomials

$$Q_i(X) = M_i(X) + \sum_j \Gamma'_{i,j}(X) F_j(X) + \sum_u \gamma'_{i,u} F'_u(X),$$

where F_j and F'_u are the parallel and aggregated constraint polynomials evaluated on P .

2: **return** $Q = (Q_1, \dots, Q_\rho)$.

Algorithm 14 PACS.PIOPOpenAndCheck(pcsCom, Q) (verifier)

- 1: Sample a query set $E' \subset S$ with $|E'| = \ell'$ (Fiat–Shamir in NIZK).
 - 2: Verify PCS openings of P and M at E' (via LVCS/DECS); reject on failure.
 - 3: Check that each $Q_i(e)$ matches its definition from $M_i(e)$ and the constraint evaluations at $e \in E'$.
 - 4: Check the masked Ω -sum test $\sum_{\omega \in \Omega} Q_i(\omega) = 0$ for all i .
 - 5: **return** accept/reject.
-

In explicit-domain mode the prover also supplies the formal coefficient descriptions of the active parallel and aggregated families. This ensures that decode compositions, theta-lifted replay rows, and aggregated bridge polynomials are replayed as polynomial identities at E' .

C.9 Fiat–Shamir compilation

SmallWood compiles the interactive PACS/PIOP protocol to a NIZK using Fiat–Shamir with grinding, following [FR25b]. In our instantiation both issuance and showing use non-empty parallel and aggregated families, so the verifier derives one batched challenge schedule and checks both the replay identities at E' and the active Ω -sum constraints.

Algorithm 15 PACS.ToNIZK (Fiat–Shamir wrapper with grinding)

- 1: Sample $\text{salt} \xleftarrow{\$} \{0, 1\}^{2\lambda}$.
 - 2: Derive challenges by hashing the transcript with grinding counters to obtain required prefix properties.
 - 3: Output PCS/LVCS/DECS openings and any required batched values; the verifier recomputes challenges and checks.
-

D EXTENDED PARAMETERS AND BENCHMARKS

This appendix provides extended parameter discussion and additional benchmark data.

D.1 Admissibility checklist

A parameter tuple is *admissible* for ARC-SPRUCE if it simultaneously satisfies:

- (1) **Bound vs. modulus:** $\beta_{\text{msg}} \ll q$ so membership checks on $[-\beta_{\text{msg}}, \beta_{\text{msg}}]$ do not wrap modulo q .
- (2) **Sampler hiding and shortness:** Gaussian widths exceed the required smoothing threshold and the signature bound β_{sig} holds with overwhelming probability.
- (3) **Rational-hash guard:** denominator invertibility succeeds with negligible failure probability under the holder randomness distribution.
- (4) **SmallWood degree/soundness:** the DECS/LVCS degree bounds and PIOP soundness parameters meet the target security λ .
- (5) **PRF entropy:** truncation length ℓ_{tag} yields at least λ bits of tag entropy.

D.2 Smoothing and sampler budgets

Let $\eta_\varepsilon(\Lambda)$ denote the smoothing parameter of a lattice Λ . A sufficient condition for statistical hiding in Gaussian preimage sampling is to choose Gaussian widths above a floor $\sigma_{\min} \geq \eta_\varepsilon(\Lambda_\perp(A))$, with slack (Appendix B). For the ambient lattice $\mathbb{Z}^{2N}$ one may use the classical bound

$$\eta_\varepsilon(\mathbb{Z}^{2N}) = \sqrt{\ln(4N(1 + 2^\lambda))}/\pi.$$

In our tuning, trapdoor quality α and acceptance parameters are chosen so that the output shortness bound β_{sig} holds with overwhelming probability.

D.3 Example lattice parameters

Table 2 records one concrete instantiation compatible with the sizing discussion in Section 6.

Knob / Result	Symbol	Value (example)
Ring degree	N	1024
Modulus	q	1038337
Security target	λ	128
Trapdoor quality	α	≈ 1.23 (measured)
Gaussian width scale	σ	tuned per-slot (Appendix B)
Signature bound	β_{sig}	derived from sampler tail bound

Table 2: Example lattice parameters (illustrative; see Appendix B for the sampling interface and tuning).

D.4 SmallWood soundness terms

SmallWood’s soundness is a combination of degree soundness (DECS), LVCS binding, and PIOP sampling errors [FR25b]. For parameters $(N, s, \ell, \ell', \rho, \eta)$ and degree bounds $d_{\text{decs}} = s + \ell - 1$ and d_Q , typical upper bounds take the form:

$$\varepsilon_{\text{decs}} \lesssim \binom{N}{d_{\text{decs}} + 2} q^{-\eta}, \quad \varepsilon_{\text{sum}} = q^{-\rho}, \quad \varepsilon_{\text{tail}} \approx \frac{\binom{d_Q}{\ell'}}{\binom{|S|}{\ell'}}.$$

Our issuance statement uses only linear constraints (given public T), so d_Q is driven by membership checks; the showing statement additionally depends on the PRF effective degree (Appendix E).

D.5 SmallWood knobs (example)

For the issuance and showing proofs discussed in Section 6, one example parameter tuple is:

$$(s, \ell, \ell', \rho, \theta, \eta) = (16, 25, 2, 2, 4, 19),$$

with masking degree d_Q derived from the instantiated constraint degrees. The pre-sign statement remains linear in the witness (given public T), while the post-sign statement's effective degree is dominated by the PRF S-box composition.

D.6 Extended benchmark grid (SmallWood)

For additional context on how SmallWood knobs affect proof size and proving time, Table 3 reports an example sweep over packing and soundness parameters (small-field variant).

ProofKB	Time (s)	Bits	NCols	ℓ	ℓ'	ρ	θ	η
9.32	0.266	133.44	6	22	2	2	4	17
10.07	0.228	146.93	4	24	2	2	4	17
11.52	0.250	133.22	4	28	2	2	4	17
14.06	0.630	198.01	4	34	8	5	2	26
14.17	0.654	192.26	6	34	8	5	2	26
14.24	0.670	198.48	4	34	8	6	2	26

Table 3: Example sweep over SmallWood knobs (proof size, prover time, and soundness bits). Sizes exclude public parameters.

E PRF DETAILS AND MISCELLANEOUS

This appendix records a concrete ZK-friendly PRF instantiation compatible with the canonical ARC protocol.

E.1 PRF specification

We use a Poseidon2-like permutation with feed-forward and truncation. Let $k \in \mathbb{F}_q^{\ell_{\text{key}}}$ be the secret key (derived from the signed $\mathbf{k}$) and $n \in \mathbb{F}_q^{\ell_{\text{nonce}}}$ encode the public nonce. Let $t = \ell_{\text{key}} + \ell_{\text{nonce}}$ be the permutation width. Define

$$F(k, n) = \text{Tr}(P(k||n) + (k||n)),$$

where $\text{Tr} : \mathbb{F}_q^t \rightarrow \mathbb{F}_q^{\ell_{\text{tag}}}$ outputs the first ℓ_{tag} lanes and P is a Poseidon2-style permutation.

E.2 Permutation structure

Let $d \geq 3$ be the smallest constant such that $\gcd(d, q - 1) = 1$. Fix round counts R_F (external rounds, even) and R_P (internal rounds), round constants, and MDS matrices M_E, M_I . The permutation has the form

$$P = E_{R_F} \circ \dots \circ E_{R_F/2+1} \circ I_{R_P} \circ \dots \circ I_1 \circ E_{R_F/2} \circ \dots \circ E_1.$$

Each external round applies the S-box to all lanes, while each internal round applies the S-box to lane 1 only. This structure reduces nonlinear constraints in the post-sign NIZK while retaining diffusion.

External and internal rounds. Let $x = (x_1, \dots, x_t) \in \mathbb{F}_q^t$. For each external round $i \in [1, R_F]$, fix per-lane constants $(c_1^{(i)}, \dots, c_t^{(i)}) \in \mathbb{F}_q^t$. For each internal round $i \in [1, R_P]$, fix a scalar constant $\hat{c}^{(i)} \in \mathbb{F}_q$. Then:

$$E_i(x) = M_E \cdot ((x_1 + c_1^{(i)})^d, \dots, (x_t + c_t^{(i)})^d)^\top,$$

$$I_i(x) = M_I \cdot ((x_1 + \hat{c}^{(i)})^d, x_2, \dots, x_t)^\top.$$

The matrices $M_E, M_I \in \mathbb{F}_q^{t \times t}$ are chosen so that they are invertible and provide sufficient diffusion (e.g., Cauchy-style MDS matrices as in Poseidon2).

E.3 Parameter generation

All PRF parameters are public. A typical parameter generator fixes:

- the base field modulus q (shared with the rest of the protocol),
- the S-box exponent d with $\gcd(d, q - 1) = 1$,
- width $t = \ell_{\text{key}} + \ell_{\text{nonce}}$ and truncation ℓ_{tag} ,
- round counts R_F, R_P ,
- MDS matrices M_E, M_I and round constants $\{c^{(i)}\}, \{\hat{c}^{(i)}\}$.

The truncation length is chosen to satisfy $\ell_{\text{tag}} \geq \lceil \lambda / \log_2(q) \rceil$, ensuring that the tag contains λ bits of entropy.

E.4 Arithmetization strategy (System A with checkpoints)

Status of the checkpoint schedule. Section 5.4 uses only the abstract interface of three PRF companion families: key binding, nonlinear checkpoint constraints, and final feed-forward/truncation constraints. The grouped schedule recorded in this appendix is the concrete realization used for the present instantiation and measurements. Alternative checkpoint schedules with the same interface could be substituted without changing the statement layer of Section 5.

To keep showing proofs compact, we commit only selected nonlinear checkpoints:

- **Key lanes.** We commit one row per key lane $\text{Key}[i]$ and enforce equality to an encoding of signed $\mathbf{k}$ using selector constraints at fixed Ω -coordinates.
- **S-box outputs.** We commit only the S-box outputs at a checkpoint schedule (e.g., every internal round and periodically among external rounds). Linear wiring between checkpoints (add-round-constants, MDS multiplications) is recomputed by the verifier from opened evaluations and public constants.

This yields a System-A style constraint system where the dominant degree comes from composing at most a small number of unchecked S-box layers between checkpoints.